

Shallow ReLU^s Networks in L^p -Type and Sobolev Spaces: Approximation and Path-Norm Controlled Generalization*

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Abstract

We study approximation by shallow ReLU^s networks, $\sigma_s(t) = \max\{0, t\}^s$, and the generalization behavior of such networks under ℓ_1 path-norm control. For the L^p -type integral spaces $\tilde{\mathcal{F}}_{p, \tau_d, s}$, $1 \leq p \leq 2$, we prove approximation bounds for shallow networks using spherical harmonic analysis. In particular, when the parameter measure is the uniform measure τ_d and $p < p^* = \frac{2d+2}{d+3}$, we obtain the rate $O\left(m^{-\frac{1}{2} - \frac{d(2-p)}{2d(2-p)+2p(2s+d+1)}} \log^{\frac{3}{2}} m\right)$, which improves the corresponding random-feature rate. We also derive approximation rates for Sobolev spaces $W^{\alpha, p}$ in the range $1 \leq p < 2$ by embedding them into spectral Barron spaces. Finally, for nonparametric regression with sub-Gaussian noise, we prove minimax-optimal generalization bounds for path-norm-regularized shallow ReLU^s networks: the rates are $O\left(n^{-\frac{d+2s+1}{2d+2s+1}} \log n\right)$ over $\mathcal{B}_s$ and $O\left(n^{-\frac{2\alpha}{2\alpha+d}} \log n\right)$ over $W^{\alpha, \infty}$, with matching lower bounds up to logarithmic factors.

Keywords: Neural Networks, Sobolev Spaces, Variation Spaces, Path Norm

1 Introduction

Deep learning has shown remarkable effectiveness in high-dimensional approximation problems, particularly in scientific computing, inverse problems, and operator learning (Han et al., 2018; Adcock et al., 2022; Beck et al., 2023). In many such settings, the ReLU^s activation $\sigma_s(t) = \max\{0, t\}^s$ ($s \in \mathbb{N}_0$) is especially relevant because it yields piecewise-polynomial representations that are well suited to smooth targets and derivative-sensitive tasks (Yang and Zhou, 2025; He et al., 2024).

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To obtain meaningful approximation guarantees, one must first identify an appropriate function class. Reproducing kernel Hilbert spaces (RKHSs) are a standard choice, but they cannot effectively approximate even a single ReLU neuron (Yehudai and Shamir, 2019). More fundamentally, Steinwart (2024) proved that on an uncountable compact space, no RKHS contains all continuous functions. This result highlights an intrinsic limitation of RKHSs: they are naturally adapted to smooth functions and therefore have limited expressive power for nonsmooth targets. Sobolev spaces $W^{\alpha,p}$, by contrast, accommodate such nonsmooth functions and are central in approximation theory and the analysis of partial differential equations (Triebel, 1983). However, Sobolev spaces are also too large from a statistical perspective: *any* machine learning method is statistically ineffective over the full Sobolev class (Wainwright, 2019). This raises a basic theoretical question: *which function classes can neural networks learn efficiently, and how does that answer depend on the activation function and the regularity of the target space?* E et al. (2022) reformulated Barron spaces (Barron, 1993) and showed that, for shallow ReLU networks, the Barron space is exactly the largest function class that can be learned statistically efficiently.¹

The connection between RKHSs and Barron spaces can be made precise through the L^p -type spaces studied by Celentano et al. (2021) and Chen et al. (2025). These works consider the following infinite-width ReLU^s model:

$$f(\mathbf{x}) = \int_{\mathcal{V}} a(\mathbf{w}, b) \sigma_s(\mathbf{w}^\top \mathbf{x} + b) d\nu(\mathbf{w}, b),$$

where $\mathcal{V} = \mathbb{S}^{d-1} \times [-1, 1]$ is the parameter space for $(\mathbf{w}, b)$ and $d\nu$ is a probability measure on $\mathcal{V}$. If $a(\mathbf{w}, b) \in L^p(d\nu)$, then f belongs to an L^p -type space, denoted by $f \in \tilde{\mathcal{F}}_{p,\nu,s}$ (with formal definition in Section 2). In particular, $\tilde{\mathcal{F}}_{p,\nu,s}$ reduces to an RKHS when $p = 2$, while the Barron space is close to a variational space (Bach, 2017) and can be identified with the union of a family of L^p -type spaces (Chen et al., 2025; Celentano et al., 2021). By Hölder’s inequality, for $1 \leq p \leq 2$,

$$\tilde{\mathcal{F}}_{2,\nu,s} \subseteq \tilde{\mathcal{F}}_{p,\nu,s} \subseteq \tilde{\mathcal{F}}_{1,\nu,s},$$

so these spaces form a natural bridge from RKHSs to Barron spaces. The connections among these spaces can be conceptually described by Figure 1. The L^p -type spaces describe the size of the neural-network integral representation, whereas Sobolev spaces describe classical smoothness. The two are connected through embedding theorems: suitable Sobolev regularity implies membership in a spectral Barron or related L^p -type representation space, which then yields shallow-network approximation rates. Precise definitions of these function spaces are given in Section 2.

We study the limits of approximating functions in L^p -type spaces and Sobolev spaces by shallow ReLU^s networks. For width m , let

$$\Sigma_m^s = \left\{ f_m = \frac{1}{m} \sum_{j=1}^m a_j \sigma_s(\mathbf{w}_j^\top \mathbf{x} + b_j) : a_j \in \mathbb{R}, \mathbf{w}_j \in \mathbb{R}^d, b_j \in \mathbb{R} \right\},$$

1. If a function belongs to the Barron space, then shallow ReLU networks can learn it without suffering from the curse of dimensionality; conversely, if a function can be sufficiently learned by shallow networks without the curse of dimensionality, then it must belong to the Barron space.

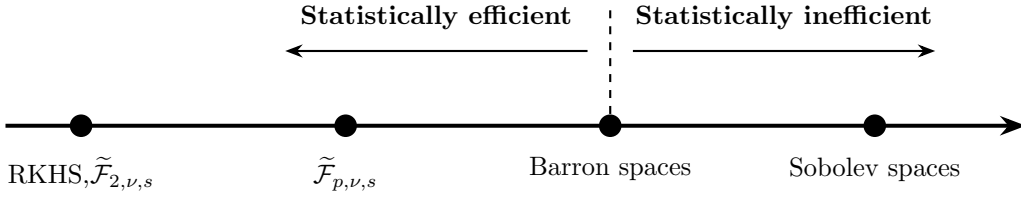


Figure 1: Hierarchy of function spaces considered in this paper, ordered from smaller to larger in generality.

and let $\Sigma^s = \bigcup_{m=1}^{\infty} \Sigma_m^s$ denote the space of shallow ReLU^s networks of arbitrary width. Studying approximation over $\tilde{\mathcal{F}}_{p,\nu,s}$ clarifies how the difficulty of learning changes along the spectrum from RKHSs to Barron spaces. For Sobolev spaces, the theory is relatively complete when $p \geq 2$, whereas the regime $1 \leq p < 2$ remains much less understood. This regime is important in PDE analysis and inverse problems with nonsmooth or singular structure: solutions with shocks, edges, or sparsity often belong naturally to $W^{\alpha,p}$ for $1 \leq p < 2$, but need not lie in the Hilbert-type space $W^{\alpha,2}$. Extending approximation guarantees to this setting is therefore relevant both theoretically and practically.

1.1 Related Work

We first discuss previous work on approximation over L^p -type spaces and Barron spaces, and then give an overview of generalization results for shallow ReLU^s networks related to such spaces.

For approximation over L^p -type spaces, the existing literature covers only the two end-point cases $p = 1$ and $p = 2$ (Bach, 2017). In the intermediate regime $1 < p < 2$, Chen et al. (2025) studied random feature models over $\tilde{\mathcal{F}}_{p,\nu,s}$ through a dual framework linking approximation and estimation, and obtained the improved approximation upper bound $O(m^{-1+\frac{1}{p}} \log^{\frac{3}{2}} m)$ for random feature models. That framework, however, does not extend directly to general shallow networks. On the other hand, Bach (2017) and Siegel (2025) studied approximation of $\tilde{\mathcal{F}}_{1,\nu,s}$ via zonoid and zonotope theory, but their arguments rely heavily on the equivalent integral representation (1), which makes extension to general p difficult.

In Sobolev approximation, a central challenge is to measure the regularity of the target in an L^q norm while deriving error bounds in an L^p norm. When $q \geq p$, optimal approximation rates are known; when $q < p$, the problem becomes substantially harder and the available results are limited. Using the embedding $W^{s+\frac{d+1}{2},2} \hookrightarrow \mathcal{B}_s$, Mao et al. (2026); Siegel and Xu (2024) established the estimate

$$\inf_{f_m \in \Sigma_m^s} \|f - f_m\|_{L^\infty} \lesssim \|f\|_{W^{s+\frac{d+1}{2},2}} m^{-\frac{1}{2} - \frac{2s+1}{2d}}.$$

This result shows that regularity measured in L^2 can still yield an optimal approximation rate in L^∞ , providing the first result in the regime $p > q$. However, it covers only the parameter setting $q = 2$, $p = \infty$, and $\alpha = s + \frac{d+1}{2}$; other regimes remain open. In particular,

when $\alpha > s + \frac{d+1}{2}$, one can derive only the upper bound $O\left(m^{-\frac{1}{2} - \frac{2s+1}{2d}}\right)$ by combining embeddings with known approximation results on $\mathcal{B}_s$. More precisely, for $2 \leq p \leq \infty$, $s \in \mathbb{N}_0$, $\alpha > 0$, and $f \in W^{\alpha,p}$, one has

$$\inf_{f_m \in \Sigma_m^s} \|f - f_m\|_{L^p} \lesssim \|f\|_{W^{\alpha,p}} \begin{cases} m^{-\frac{1}{2} - \frac{2s+1}{2d}}, & \alpha \geq s + \frac{d+1}{2}, \\ m^{-\frac{\alpha}{d}}, & \alpha \in \mathbb{N} \cap (0, s + \frac{d+1}{2}). \end{cases}$$

When $\alpha < s + \frac{d+1}{2}$, achieving the optimal rate requires α to be an integer, and the noninteger case remains open. Moreover, the available theory applies only to $2 \leq p \leq \infty$, mainly because Mao et al. (2026) relies on the spectral Barron space $\mathcal{B}_s$ as an intermediate space, while the embedding $W^{\alpha,p} \hookrightarrow \mathcal{B}_s$ generally fails for $1 \leq p < 2$. We address this regime by establishing new embeddings between Sobolev spaces and spectral Barron spaces.

There is also substantial literature on the generalization behavior of shallow ReLU^s networks. Bach (2017) and Parhi and Nowak (2023) systematically analyzed infinite-width ReLU models. In particular, Bach (2017) proved a representation theorem for integral ReLU networks with variational-norm regularization, implying that a finite-width network f_m can always be trained, and established the generalization upper bound $O\left(n^{-\frac{1}{2}}\right)$ over the variational space $\tilde{\mathcal{F}}_{1,\nu,1}$. Parhi and Nowak (2023) used Barron regularization to show that the minimax-optimal rate over Barron spaces is $O\left(n^{-\frac{d+3}{2d+3}}\right)$. For finite-width networks, Liu et al. (2024) explored the trade-off between sample complexity and dimension dependence, and obtained the upper bound $O\left(n^{-\frac{d+2}{2d+2}}\right)$ over Barron spaces under ℓ_1 path norm constraints, although that rate is not minimax optimal but the dimension dependence is relaxed. By contrast, Yang and Zhou (2024) considered ℓ_2 path norms and proved minimax-optimal rates $O\left(n^{-\frac{d+3}{2d+3}}\right)$ and $O\left(n^{-\frac{2\alpha}{2\alpha+d}}\right)$ over Barron and Hölder spaces, respectively. Since the ℓ_2 path norm leads to a formulation equivalent to $\|\cdot\|_2$ -regularized learning (Yang and Zhou, 2024), its analysis is substantially cleaner than that of the ℓ_1 counterpart.

1.2 Contributions

Our results focus on shallow ReLU^s networks. We prove approximation results over L^p -type spaces $\tilde{\mathcal{F}}_{p,\nu,s}$, extend the Sobolev approximation theory to $W^{\alpha,p}$ when $1 < p < 2$, and establish minimax-optimal generalization bounds over Barron and Sobolev spaces. The main contributions are as follows:

- Using spherical harmonic analysis, we establish new approximation results for shallow ReLU^s networks over L^p -type spaces $\tilde{\mathcal{F}}_{p,\nu,s}$. In particular, when the parameter measure is the uniform distribution τ_d and $p < p^* = \frac{2d+2}{d+3}$, the approximation rate is $O\left(m^{-\frac{1}{2} - \frac{d(2-p)}{2d(2-p)+2p(2s+d+1)}} \log^{\frac{3}{2}} m\right)$, improving upon the rate obtained by random feature models in Chen et al. (2025). Thus, optimizing the inner parameters of a shallow network can improve approximation over random feature models for relatively harder target classes.
- By establishing embeddings between spectral Barron spaces and Sobolev spaces, we derive approximation rates for shallow ReLU^s networks over $W^{\alpha,p}$ when $1 < p < 2$.

In particular, when $\alpha \geq s + d$, the rate is $O\left(m^{-\frac{1}{2} - \frac{s-1}{d+1}} \log m\right)$; when $\alpha \in \mathbb{N} \cap (0, s + d)$, the rate is $O\left(m^{-\frac{\alpha(2s+d+3)}{2(s+d)(d+1)}} \log m\right)$.

- We prove that for any $s \in \mathbb{N}_0$, shallow ReLU^s networks with ℓ_1 path norm constraints achieve minimax-optimal generalization bounds over both Barron and Sobolev spaces. Our analysis builds upon techniques from (Yang and Zhou, 2024; Wainwright, 2019). The framework of Yang and Zhou (2024) applies only to bounded or Gaussian noise; here we extend the generalization bounds to sub-Gaussian noise distributions using generic chaining.

The rest of the paper is organized as follows. Section 2 reviews the shallow-network model and the relevant Barron, L^p -type, and Sobolev spaces. Section 3 presents the main approximation results. Section 4 establishes minimax-optimal generalization bounds. Section 5 concludes the paper. Appendix A collects the notation used throughout.

2 Preliminaries

We recall the function spaces and learning framework used in the approximation and generalization results.

2.1 Function Spaces

Barron spaces Consider functions of the following form:

$$f_\rho(\mathbf{x}) = \mathbb{E}_{(a, \mathbf{w}, b) \sim \rho} \left[a \sigma_s \left(\mathbf{w}^\top \mathbf{x} + b \right) \right], \quad \mathbf{x} \in \Omega,$$

where $d\rho$ is a probability measure on the Borel σ -algebra. Such functions are precisely infinite-width shallow ReLU^s networks. Define the *sth-order Barron space* by

$$\mathcal{B}_s = \{f = f_\rho : \rho \text{ is a probability measure}\},$$

and define the *sth-order Barron norm* of f_ρ by

$$\|f_\rho\|_{\mathcal{B}_s} = \inf \left\{ \mathbb{E}_{(a, \mathbf{w}, b) \sim \rho} [|a| (\|\mathbf{w}\|_1 + |b|)^s] : f = f_\rho \right\}.$$

Here, $\mathcal{B}_1$ is exactly the Barron space introduced in E et al. (2022), for which optimal approximation rates and metric-entropy estimates are already known (Siegel and Xu, 2024). For general $s \in \mathbb{N}_0$, one can show that $\mathcal{B}_s$ is equivalent to the variation space defined in Siegel and Xu (2024); as a consequence, the same approximation and entropy estimates remain available. This equivalence is one byproduct of the present work.

Lemma 1 *Let $s \in \mathbb{N}_0$. Then for any $f \in \mathcal{B}_s$, one has*

$$\inf_{f_m \in \Sigma_m^s} \|f - f_m\|_{L^\infty} \lesssim \|f\|_{\mathcal{B}_s} m^{-\frac{1}{2} - \frac{2s+1}{2d}}.$$

The proof of this lemma relies on the equivalence between the Barron space and the variation space, which allows us to invoke the corresponding known results directly. Details are given in Appendix B.

L^p -type spaces The study of networks in integral form can be traced back to kernel methods and random feature models. If a signed Radon measure admits a density $a(\mathbf{w}, b)$ with respect to a fixed probability measure ν , then the collection of all functions satisfying

$$\int_{\mathcal{V}} |a(\mathbf{w}, b)|^2 d\nu(\mathbf{w}, b) < \infty$$

forms a reproducing kernel Hilbert space (RKHS) (Bach, 2017; Celentano et al., 2021; Chen et al., 2025), with reproducing kernel

$$k(\mathbf{x}, \mathbf{x}') = \int_{\mathcal{V}} \sigma_s(\mathbf{w}^\top \mathbf{x} + b) \sigma_s(\mathbf{w}^\top \mathbf{x}' + b) d\nu(\mathbf{w}, b).$$

Although kernel methods are mathematically well developed, they suffer from the curse of dimensionality. For example, they struggle to learn ridge functions that depend on only a few input coordinates. To mitigate this limitation, one can generalize the RKHS construction. Since an RKHS is essentially a weighted L^2 space, Bach (2017) introduced the corresponding weighted L^1 space in the ReLU setting, namely the class of functions satisfying $\int_{\mathcal{V}} |a(\mathbf{w}, b)| d\nu(\mathbf{w}, b) < \infty$, often called convex neural networks. Celentano et al. (2021); Chen et al. (2025) then considered the general case $p \geq 1$, leading to the family of L^p -type spaces

$$\tilde{\mathcal{F}}_{p,\nu,s} = \left\{ f(\mathbf{x}) = \int_{\mathcal{V}} a(\mathbf{w}, b) \sigma_s(\mathbf{w}^\top \mathbf{x} + b) d\nu(\mathbf{w}, b) : a(\mathbf{w}, b) \in L^p(d\nu) \right\},$$

equipped with the norm

$$\|f\|_{\tilde{\mathcal{F}}_{p,\nu,s}} = \inf \left\{ \|a(\mathbf{w}, b)\|_{L^p(d\nu)} : f(\mathbf{x}) = \int_{\mathcal{V}} a(\mathbf{w}, b) \sigma_s(\mathbf{w}^\top \mathbf{x} + b) d\nu(\mathbf{w}, b) \right\}.$$

For a finite network

$$f_m(\mathbf{x}) = \frac{1}{m} \sum_{j=1}^m a_j \sigma_s(\mathbf{w}_j^\top \mathbf{x} + b_j),$$

the corresponding discrete control is the generalized ℓ_1 path norm

$$\|f_m\|_{\mathcal{P}_{1,s}} = \frac{1}{m} \sum_{j=1}^m |a_j| (\|\mathbf{w}_j\|_1 + |b_j|)^s.$$

This quantity is the atomic, finite-width analogue of the Barron or variation norm: replacing the finite sum by a signed measure gives the integral representation underlying Barron spaces. This link motivates the path-norm control used in the generalization analysis below.

Sobolev spaces Sobolev spaces play a central role in areas such as partial differential equations. We use the standard definition based on weak derivatives. For $1 \leq p \leq \infty$, the *Sobolev space* of order α is defined by

$$W^{\alpha,p}(\mathbb{R}^d) = \left\{ f \in L^p(\mathbb{R}^d) : \|f\|_{W^{\alpha,p}(\mathbb{R}^d)} < \infty \right\},$$

where:

1. When α is an integer,

$$\|f\|_{W^{\alpha,p}(\mathbb{R}^d)} = \sum_{|\beta| \leq \alpha} \|\partial^\beta f\|_{L^p(\mathbb{R}^d)};$$

2. When α is not an integer, write $\alpha = k + \theta$, where k is a nonnegative integer and $\theta \in (0, 1)$. Then

$$\|f\|_{W^{\alpha,p}(\mathbb{R}^d)} = \sum_{|\beta| \leq k} \|\partial^\beta f\|_{L^p(\mathbb{R}^d)} + \sum_{|\beta|=k} \left\| \frac{\partial^\beta f(x) - \partial^\beta f(y)}{\|x - y\|_2^{\frac{d}{p} + \theta}} \right\|_{L^p(\mathbb{R}^d \times \mathbb{R}^d)}.$$

Restricting $W^{\alpha,p}(\mathbb{R}^d)$ to Ω yields the corresponding restricted Sobolev space $W^{\alpha,p}(\Omega)$.

2.2 Background: Generalization Analysis

Unless otherwise specified, the input domain is $\Omega = \mathbb{B}^d$ (all results can in fact be extended to general compact sets with Lipschitz boundary), and the label space is $Z \subseteq \mathbb{R}$. Let (X, Y) follow the true distribution $d\rho$ on $\Omega \times Z$, and suppose that we observe n i.i.d. noisy samples $D_n = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$. The goal of the *regression problem* is to learn, from D_n , a function $f_m \in \Sigma^s$ that approximates the regression function

$$f_\rho(\mathbf{x}) = \mathbb{E}[Y \mid X = \mathbf{x}] = \int_Y y \, d\rho(y \mid \mathbf{x}), \quad \mathbf{x} \in \Omega,$$

where $\rho(\cdot \mid \mathbf{x})$ denotes the conditional distribution of Y given $X = \mathbf{x}$. It is standard to assume that f_ρ belongs to a hypothesis space $\mathcal{H}$, such as a Barron space (Liu et al., 2024) or a Sobolev space (Yang and Zhou, 2024, 2025), reflecting prior knowledge about the target.

Besides fitting the data well, neural-network training must also control model complexity in order to ensure good generalization. In the overparameterized regime, the classical viewpoint that measures complexity by parameter count is no longer adequate; the double-descent phenomenon (Belkin et al., 2020; Hastie et al., 2022) is a standard example. This has motivated a norm-based view of training, in which the optimization procedure is understood as explicitly or implicitly minimizing a suitable weight norm, often called a representation cost (Neyshabur et al., 2015; Ongie et al., 2020; Liu et al., 2024; Yang and Zhou, 2024).

We adopt explicit regularization and consider the following L^2 empirical risk minimization problem:

$$f_m \in \arg \min_{f \in \Sigma^s} \left\{ \frac{1}{n} \sum_{i=1}^n |f(\mathbf{x}_i) - y_i|^2 + \lambda \|f\| \right\},$$

where $\lambda \equiv \lambda(n) > 0$ is the regularization parameter and $\|\cdot\|$ denotes a weight norm of the network. We choose the (generalized) ℓ_1 path norm (E et al., 2022; Liu et al., 2024):

$$\|f\|_{\mathcal{P}_{1,s}} = \frac{1}{m} \sum_{i=1}^m |a_i| (\|\mathbf{w}_i\|_1 + |b_i|)^s,$$

which can be viewed as a discrete analogue of the Barron norm. Denote by $\Sigma_{m,M}^s$ the set of shallow ReLU^s networks with m neurons and ℓ_1 path norm at most M . Note that the path norm is not parametrization invariant: two different parameterizations of the same network function may have different path norms. This dependence on representation is precisely why the path norm provides a meaningful measure of model complexity.

We consider the expected generalization error of the estimator f_m ,

$$\mathbb{E}_{D_n} \left[\|f_m - f_\rho\|_{L^2(d\mu)} \right],$$

where $d\mu$ is the marginal law of X . If one can prove an upper bound together with a matching minimax lower bound, then the estimator is statistically optimal.

3 Approximation Results

We now present approximation results for shallow ReLU^s networks over the L^p -type space $\tilde{\mathcal{F}}_{p,\tau_d,s}$ and the Sobolev space $W^{\alpha,p}$. The first result treats L^p -type integral spaces with optimized inner parameters. The second concerns Sobolev spaces in the range $1 \leq p < 2$, where Hilbert-space methods based on L^2 regularity no longer apply directly.

3.1 Approximation of L^p -type Spaces

We first study approximation of the integral representation class $\tilde{\mathcal{F}}_{p,\tau_d,s}$, where the parameter measure is the uniform measure τ_d on the sphere. Previous results for the intermediate range $1 < p < 2$ were mainly obtained for random feature models, in which the inner parameters are sampled and then fixed. Here the inner parameters are optimized as part of the shallow network, and this additional flexibility leads to a better approximation rate when p is below the critical index p^* .

Theorem 2 (Approximation of L^p -type spaces by shallow networks) *Let τ_d be the uniform measure on $\mathbb{S}^d$. For $d \geq 2$, $s \in \mathbb{N}_0$, $1 \leq p \leq 2$, and $f \in \tilde{\mathcal{F}}_{p,\tau_d,s}$, the following bound holds for every $m \in \mathbb{N}$:*

$$\inf_{f_m \in \Sigma_m^s} \|f_m - f\|_{L^2} \lesssim \begin{cases} m^{-\frac{1}{2} - \frac{d(2-p)}{2d(2-p)+2p(2s+d+1)}} \log^{\frac{3}{2}} m, & 1 \leq p < p^*, \\ m^{-\frac{1}{2}} \log^2 m, & p = p^*, \\ m^{-\frac{p(2s+d+1)}{p(d+3)-2(d+1)+2p(2s+d+1)}} \log^{\frac{3}{2}} m, & p^* < p \leq 2, \end{cases}$$

where $p^* = \frac{2d+2}{d+3}$.

The proof of Theorem 2 is based on an intermediate approximation step. Instead of approximating $f \in \tilde{\mathcal{F}}_{p,\tau_d,s}$ directly by a finite network, we first approximate it by a function in the Hilbert-type space $\tilde{\mathcal{F}}_{2,\tau_d,s}$ and then approximate this intermediate function by a shallow network. The first step is where the L^p structure enters.

We begin on the sphere. After writing the parameter as $\boldsymbol{\theta} = (\mathbf{w}^\top, b) \in \mathbb{S}^d$, the uniform measure τ_d allows us to diagonalize the integral operator associated with $\sigma_s(\boldsymbol{\theta}^\top \mathbf{u})$ by spherical harmonics. This gives the following approximation result.

Lemma 3 (Approximation on the unit sphere) *Let $d \geq 2$, $s \in \mathbb{N}_0$, $1 \leq p \leq 2$, and $\tilde{f} \in \tilde{\mathcal{F}}_{p,\tau_d,s}(\mathbb{S}^d)$. Assume that $\tilde{f}$ has parity opposite to that of s . Then, for every $R > 0$,*

$$\inf_{\|g\|_{\tilde{\mathcal{F}}_{2,\tau_d,s}(\mathbb{S}^d)} \leq R} \|\tilde{f} - g\|_{L^2(\mathbb{S}^d)} \lesssim \|\tilde{f}\|_{\tilde{\mathcal{F}}_{p,\tau_d,s}(\mathbb{S}^d)} \begin{cases} R^{-\frac{2s+d+1}{d} \cdot \frac{p}{2-p}}, & 1 \leq p < p^*, \\ Re^{-(s+\frac{d+1}{2})R^2}, & p = p^*, \\ R^{-\frac{2p(2s+d+1)}{p(d+3)-2(d+1)}}, & p^* < p \leq 2, \end{cases}$$

where $p^* = \frac{2d+2}{d+3}$.

The key point of Lemma 3 is that the L^p integrability of the outer coefficient $a(\boldsymbol{\theta})$ can be converted into decay of the spherical-harmonic components of $\tilde{f}$. The boundedness range of the spherical-harmonic projection operators is exactly what produces the critical index p^* .

The next step transfers the spherical approximation back to the original domain. We use the lifting

$$\tilde{f}(\mathbf{u}) = u_{d+1}^s f\left(\frac{\mathbf{u}'}{u_{d+1}}\right), \quad \mathbf{u} = (\mathbf{u}', u_{d+1}) \in \mathbb{S}^d, \quad u_{d+1} > 0,$$

and then restrict the spherical approximant back to Ω .

Corollary 4 (From $\mathbb{S}^d$ to $\mathbb{B}^d$) *Let $d \geq 2$, $s \in \mathbb{N}_0$, $1 \leq p \leq 2$, and $f \in \tilde{\mathcal{F}}_{p,\tau_d,s}$. Then, for every $R > 0$,*

$$\inf_{\|g\|_{\tilde{\mathcal{F}}_{2,\tau_d,s}(\mathbb{B}^d)} \leq R} \|f - g\|_{L^2} \lesssim \|f\|_{\tilde{\mathcal{F}}_{p,\tau_d,s}(\mathbb{B}^d)} \begin{cases} R^{-\frac{2s+d+1}{d} \cdot \frac{p}{2-p}}, & 1 \leq p < p^*, \\ Re^{-(s+\frac{d+1}{2})R^2}, & p = p^*, \\ R^{-\frac{2p(2s+d+1)}{p(d+3)-2(d+1)}}, & p^* < p \leq 2, \end{cases}$$

where $p^* = \frac{2d+2}{d+3}$.

Combining Corollary 4 with the available approximation bound for functions in $\tilde{\mathcal{F}}_{2,\tau_d,s}$ gives the final network approximation. Since a random-feature model is a special shallow network, (Chen et al., 2025, Theorem 6.3) implies that, if $\|g\|_{\tilde{\mathcal{F}}_{2,\tau_d,s}} \leq R$, then there exists $f_m \in \Sigma_m^s$ satisfying

$$\|g - f_m\|_{L^2} \lesssim R m^{-\frac{1}{2}} \log^{\frac{3}{2}} m.$$

Balancing this term with the intermediate error in Corollary 4 gives the three cases in Theorem 2. The proofs of Lemma 3, Corollary 4, and the final balancing argument are given in Appendix C.

The proof of Theorem 2 uses the special structure of the uniform measure τ_d in Lemma 3. For a general measure τ , the Funk–Hecke formula is no longer available, so one cannot derive a spherical-harmonic expansion analogous to (4). Moreover, even when $d\nu \ll d\tau_d$, the Radon–Nikodym derivative need not inherit the integrability of $a(\mathbf{w}, b)$, so the proof strategy breaks down. The critical value p^* is also tied to the boundedness of projection operators: the argument requires estimates for the modulus of smoothness on L^p -type spaces, and the L^p boundedness of spherical-harmonic projection operators changes at a

critical index (Kwon and Lee, 2018), producing the logarithmic factors in the bound. We conjecture that this phase transition is intrinsic rather than an artifact of the proof.

Theorem 2 shows that, away from the critical point p^* , the approximation rate depends monotonically on p : larger p leads to better approximation over $\tilde{\mathcal{F}}_{p,\tau_d,s}$. When $p = 1$, the rate over $\tilde{\mathcal{F}}_{1,\tau_d,s}$ is $O\left(m^{-\frac{1}{2}-\frac{d}{4d+4s+2}} \log^{\frac{3}{2}} m\right)$; letting $d \rightarrow \infty$ yields the dimension-independent bound $O\left(m^{-\frac{3}{4}} \log^{\frac{3}{2}} m\right)$, which is better than the known rate over the Barron space $\mathcal{B}_s$. This is consistent with the inclusion relation $\mathcal{B}_s = \bigcup_{\nu} \tilde{\mathcal{F}}_{p,\nu,s}$, and suggests that $\mathcal{B}_s$ is strictly larger than any fixed L^1 -type space. When $p = 2$, $\tilde{\mathcal{F}}_{2,\tau_d,s}$ reduces to an RKHS, and the rate becomes $O\left(m^{-\frac{1}{2}} \log^{\frac{3}{2}} m\right)$ as $d \rightarrow \infty$, matching the rate for random feature models. Although shallow networks form a strictly larger class than random feature models and might therefore seem capable of better approximation, the Monte Carlo lower bound of order $m^{-\frac{1}{2}}$ (Lin, 2017) shows that no further improvement is possible in this case. At the critical point $p = p^*$, the rate is again $O\left(m^{-\frac{1}{2}} \log^2 m\right)$, indicating that shallow networks have essentially the same approximation power over the L^{p^*} -type space as over an RKHS.

3.2 Approximation of Sobolev Spaces

We next turn to Sobolev spaces. Existing shallow-network approximation results for Sobolev functions often pass through embeddings into Barron or spectral Barron spaces, but the available embeddings are most effective when the smoothness is measured in L^2 or stronger norms. For $1 \leq p < 2$, such an embedding is more delicate. The following theorem gives an approximation bound by first embedding $W^{\alpha,p}$ into a spectral Barron space and then applying known spectral-Barron approximation estimates.

Theorem 5 (Approximation of Sobolev spaces by shallow networks) *Let $s \in \mathbb{N}_0$, $\alpha > 0$, $1 \leq p < 2$, and $f \in W^{\alpha,p}$. Then*

$$\inf_{f_m \in \Sigma_m^s} \|f - f_m\|_{L^p} \lesssim \|f\|_{W^{\alpha,p}} \begin{cases} m^{-\frac{1}{2}-\frac{s-1}{d+1}} \log m, & \alpha \geq s + d, \\ m^{-\frac{\alpha(2s+d+3)}{2(s+d)(d+1)}} \log m, & \alpha \in \mathbb{N} \cap (0, s + d). \end{cases}$$

Theorem 5 addresses the regime $1 \leq p < 2$, which is not covered by Mao et al. (2026). Its proof is based on a new embedding from the Sobolev space $W^{\alpha,p}$ into the spectral Barron space, which reduces the problem to known approximation results for the latter. Since the currently available approximation rate over the corresponding spectral Barron space is not sharp, the bound in this theorem is not expected to be optimal either. The proof is given in Appendix D.

4 Generalization Analysis

The Barron norm is equivalent to the variation norm of infinite-width ReLU representations, while the ℓ_1 path norm is its finite-width, parameter-level counterpart. This connection provides a natural way to control the complexity of trained networks through their parameters and motivates the path-norm regularization used below. We consider empirical risk mini-

mization with an ℓ_1 path-regularization term:

$$f_m \in \operatorname{argmin}_{f_\theta \in \Sigma_m^s} \left\{ \frac{1}{n} \sum_{i=1}^n |f_\theta(\mathbf{x}_i) - y_i|^2 + \lambda \|\theta\|_{\mathcal{P}_{1,s}} \right\}.$$

For simplicity, we assume throughout that the above optimization problem admits a solution.

4.1 Upper Bounds for the Generalization Error

We derive upper bounds for the generalization error of shallow ReLU^s networks under suitable regularity and noise assumptions. We begin with the following two assumptions:

Assumption 6 (Regularity assumption) *The regression function satisfies*

$$f_\rho \in W^{\alpha, \infty}(\Omega), \quad \|f_\rho\|_{W^{\alpha, \infty}(\Omega)} \leq 1,$$

where $\alpha > 0$ and, when $\alpha < s + \frac{d+1}{2}$, α is an integer.

Assumption 7 (Noise assumption) *Let $\boldsymbol{\eta} = Y - f_\rho(X)$ denote the (zero-mean) noise. Then $\boldsymbol{\eta}$ is sub-Gaussian, that is, there exists $\sigma > 0$ such that*

$$\mathbb{E} \left[e^{\lambda \boldsymbol{\eta}} \right] \leq e^{\frac{\lambda^2 \sigma^2}{2}}, \quad \forall \lambda \in \mathbb{R}.$$

Moreover, it is independent of X .

In Assumption 6, the requirement $\alpha \in \mathbb{N}$ when $\alpha < s + \frac{d+1}{2}$ comes from the approximation theorem used later. Assumption 7 is more general than the commonly used bounded-noise assumption: it includes bounded, Gaussian, and more general sub-Gaussian noise. An even broader condition is the sub-exponential assumption, namely, that there exist $R > 0$ and $C > 1$ such that

$$\mathbb{E}[|\boldsymbol{\eta}|^p | X = \mathbf{x}] \leq Cp!R^p, \quad \forall p \in \mathbb{N}, x \in \Omega,$$

for which analyses of unbounded errors can be found in (Wainwright, 2019; Guo and Zhou, 2013; Adamczak, 2008). Both the sub-Gaussian and sub-exponential conditions can be unified by requiring the Orlicz norm $\|\cdot\|_{\psi_p}$ to be finite. It is defined by

$$\|X\|_{\psi_p} = \inf\{t > 0 : \mathbb{E}[\exp(|X|^p/t^p)] \leq 2\}, \quad p \geq 1.$$

When $p = 1$ and $p = 2$, this reduces to the sub-exponential and sub-Gaussian cases, respectively. The main result of this subsection is the following theorem, proved in Appendix E.

Theorem 8 *Under Assumptions 7 and 6, consider the empirical risk minimizer*

$$f_n \in \arg \min_{f \in \Sigma_m^s} \left\{ \frac{1}{n} \sum_{i=1}^n (f(\mathbf{x}_i) - y_i)^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} \right\}.$$

Then there exist constants $c_1, c_2 > 0$ such that:

1. If $\alpha \geq s + \frac{d+1}{2}$, then by taking

$$m \gtrsim n^{\frac{d}{2d+2s+1}}, \quad \lambda \asymp n^{-\frac{d+2s+1}{2d+2s+1}} \log n$$

one has

$$\|f_n - f_\rho\|_{L^2(\mathrm{d}\mu)}^2 \lesssim n^{-\frac{d+2s+1}{2d+2s+1}} \log n$$

with probability at least $1 - c_1 e^{-c_2 n^{\frac{d}{2d+2s+1}} \log n}$.

2. If $\alpha \in \mathbb{N} \cap (0, s + \frac{d+1}{2})$, then by taking

$$m \gtrsim n^{\frac{d}{2\alpha+d}}, \quad \lambda \asymp n^{-\frac{1}{2} - \frac{2s+1}{4\alpha+2d}} \log n$$

one has

$$\|f_n - f_\rho\|_{L^2(\mathrm{d}\mu)}^2 \lesssim n^{-\frac{2\alpha}{2\alpha+d}} \log n$$

with probability at least $1 - c_1 e^{-c_2 n^{\frac{d}{2\alpha+d}} \log n}$.

4.2 Minimax Lower Bounds

The goal of this subsection is to show that the upper bound in Theorem 8 is minimax optimal. We continue to consider i.i.d. samples $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ drawn from an unknown distribution $\mathrm{d}\rho$, and estimators of the form

$$f_m \in \arg \min_{f \in \Sigma_m^s} \left\{ \frac{1}{n} \sum_{i=1}^n (f(\mathbf{x}_i) - y_i)^2 + \lambda \|f\|_{\mathcal{D}_{1,s}} \right\}.$$

For the lower-bound argument, we impose the following two assumptions:

Assumption 9 (Noise assumption) *The samples satisfy*

$$y_i = f_\rho(\mathbf{x}_i) + \eta_i, \quad \eta_i \sim \mathcal{N}(0, \sigma^2).$$

Here $\sigma^2 > 0$ is the variance.

Assumption 10 (Sample distribution assumption) *Assume that the marginal distribution of X is the uniform distribution on Ω , namely, $\mu = U(\Omega)$.*

Assumption 9 is a special case of Assumption 7. We impose Gaussian noise mainly because it simplifies the computation of the Kullback–Leibler divergence. Similar arguments should also apply to certain other sub-Gaussian distributions, such as log-concave or light-tailed laws. Assumption 10 is introduced because we need to estimate the metric entropy $\log \mathcal{N}(\varepsilon, W^{\alpha,\infty}(\Omega), \|\cdot\|_{L^2(\mathrm{d}\mu)})$. If $\mu = U(\Omega)$, then

$$\log \mathcal{N}(\delta, BW^{\alpha,\infty}(\Omega), \|\cdot\|_{L^2(\mathrm{d}\mu)}) \asymp \log \mathcal{N}(\delta, BW^{\alpha,\infty}(\Omega), \|\cdot\|_{L^2(\Omega)}).$$

In fact, it is enough to assume that $\mu \ll m$ and that $w = \frac{d\mu}{dx}$ is bounded, where m denotes Lebesgue measure; under this condition, the same conclusion continues to hold.

Given a target function class $\mathcal{F}$, we define the *minimax risk* by

$$\inf_{\hat{f}} \sup_{f_\rho \in \mathcal{F}} \mathbb{E} \left[\|\hat{f} - f_\rho\|_{L^2(d\mu)}^2 \right],$$

where the infimum is taken over all estimators, that is, all measurable functions of the sample $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$. We are interested in lower bounds of order $n^{-\beta}$. If such a lower bound matches the upper bound in Theorem 8, then the estimator f_m is statistically optimal.

Theorem 11 *Let $\alpha > 0$. Then under Assumptions 9 and 10, the minimax risk satisfies*

$$\inf_{\hat{f}} \sup_{\substack{f_\rho \in W^{\alpha, \infty}(\Omega), \\ \|f_\rho\|_{W^{\alpha, \infty}(\Omega)} \leq 1}} \mathbb{E} \left[\|\hat{f} - f_\rho\|_{L^2(d\mu)}^2 \right] \gtrsim n^{-\frac{2\alpha}{2\alpha+d}}.$$

It follows that when $\alpha < s + \frac{d+1}{2}$, the upper bound in Theorem 8 matches the minimax lower bound up to logarithmic factors, so the estimator f_n is statistically optimal. By contrast, when $\alpha \geq s + \frac{d+1}{2}$, the upper bound no longer matches the Sobolev minimax lower bound. The reason is that our upper-bound argument uses the embedding $W^{\alpha, \infty}(\Omega) \subset \mathcal{B}_s(\Omega)$, which effectively reduces the problem to the Barron class and yields the rate $n^{-\frac{d+2s+1}{2d+2s+1}} \log n$. The next theorem shows that this rate is itself minimax optimal over $\mathcal{B}_s(\Omega)$.

Theorem 12 *Under Assumptions 9 and 10, the minimax risk satisfies*

$$\inf_{\hat{f}} \sup_{\substack{f_\rho \in \mathcal{B}_s(\Omega), \\ \|f_\rho\|_{\mathcal{B}_s(\Omega)} \leq 1}} \mathbb{E} \left[\|\hat{f} - f_\rho\|_{L^2(d\mu)}^2 \right] \gtrsim n^{-\frac{d+2s+1}{2d+2s+1}}.$$

5 Conclusion and Discussion

We studied approximation and generalization for shallow ReLU^s networks. For the L^p -type space $\tilde{\mathcal{F}}_{p, \tau_d, s}$, $1 \leq p \leq 2$, spherical harmonic analysis gives approximation rates for shallow ReLU^s networks (Theorem 2), with a phase transition at $p^* = \frac{2d+2}{d+3}$. For Sobolev spaces $W^{\alpha, p}$ with $1 \leq p < 2$, we obtained L^p approximation rates (Theorem 5) through embeddings into spectral Barron spaces, covering a regime not treated in Mao et al. (2026).

On the generalization side, we analyzed nonparametric regression with ℓ_1 path-norm-regularized shallow ReLU^s networks under sub-Gaussian noise, using generic chaining. When the regression function f_ρ belongs to the Barron space $\mathcal{B}_s$, the generalization error satisfies the upper bound $O\left(n^{-\frac{d+2s+1}{2d+2s+1}} \log n\right)$; when $f_\rho \in W^{\alpha, \infty}$, it satisfies the upper bound $O\left(n^{-\frac{2\alpha}{2\alpha+d}} \log n\right)$. Combined with the corresponding minimax lower bounds (Theorems 11 and 12), these rates are minimax optimal up to logarithmic factors in their respective settings.

Several open problems remain.

- **Optimality of approximation rates in Sobolev spaces.** The rates in Theorem 5 for $1 \leq p < 2$ are not yet optimal. The main obstruction is that the currently available approximation rate over the corresponding spectral Barron space is itself not sharp. It would be interesting to develop more refined harmonic-analysis tools, such as Littlewood–Paley decompositions, to analyze approximation in $W^{\alpha,p}$ directly and possibly obtain optimal rates.
- **Approximation under general parameter measures.** The approximation results for L^p -type spaces in Theorem 2 rely on taking the parameter measure ν to be the uniform distribution τ_d , which makes the Funk–Hecke formula available. For general probability measures ν , the present proof no longer applies. Extending the theory to broader classes of parameter measures remains an open problem.
- **Approximation lower bounds.** The present work establishes only upper bounds for approximation over L^p -type spaces. Matching lower bounds, and hence minimax-optimal approximation results, remain open. Deriving such lower bounds would complete the picture behind Theorem 2.

The techniques developed here may also be useful for studying deeper architectures, where the interaction between representation norms, approximation, and statistical complexity remains less understood.

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Appendix A. Symbol Table

Symbol	Meaning	Def. page
$A \lesssim B$	There exists a constant $C > 0$ such that $A \leq CB$	–
$A \gtrsim B$	There exists a constant $C > 0$ such that $A \geq C^{-1}B$	–
$A \asymp B$	Both $A \lesssim B$ and $A \gtrsim B$ hold	–
$O(\cdot)$	Big O notation	–
$L^p(d\nu)$	L^p space with respect to the measure $d\nu$	–
$\ \cdot\ _{L^p(d\mu)}$	Norm on the space $L^p(d\mu)$	–
$\mathcal{F}(\Omega)$	Function space $\mathcal{F}$ with domain Ω explicitly indicated	–
$\mathcal{F}(R)$	Unit ball of radius R in the function space $\mathcal{F}$	–
$\mathbb{B}^d$	$\{x \in \mathbb{R}^d : \ x\ _2 \leq 1\}$ unit ball	–
$\mathbb{S}^{d-1}$	$\{x \in \mathbb{R}^d : \ x\ _2 = 1\}$ unit sphere	–
$\pi_B f$	Truncated function	30
$\pi_B(\mathcal{F})$	Family of truncated functions	28
Σ_m^s	Class of shallow networks with m neurons and ReLU ^s activation	2
$\mathcal{B}_s$	Barron space of order s	5
$\ \cdot\ _{\mathcal{B}_s}$	Barron norm of order s	5
$\tilde{\mathcal{F}}_{p,\nu,s}$	Variation space	6
$\ \cdot\ _{\tilde{\mathcal{F}}_{p,\nu,s}}$	Variation space norm	6
$W^{s,p}(\Omega)$	Sobolev space of order s	6
$\ \cdot\ _{W^{s,p}}$	Sobolev norm	7
$\mathcal{N}(\varepsilon, \mathcal{F}, \ \cdot\)$	ε -covering number	28
$\mathcal{G}_n(\mathcal{F}, \delta, \xi)$	Local complexity	27
$\text{star}(\mathcal{F})$	Star-shaped hull	27
$\gamma_2(T, d)$	Generic chaining complexity	30
$\sigma_s(\cdot)$	ReLU ^s activation function	1
$\mathcal{P}_{1,s}$	Path norm of order s	6
τ_d	Uniform measure on $\mathbb{S}^d$	18
ω_d	Surface area of $\mathbb{S}^d$	19
χ_A	Indicator function of set A	–
δ_x	Dirac measure at x	–
$\mathcal{N}(\cdot)$	Covering number	–
$\mathcal{N}_{KL}(\varepsilon; \mathcal{P})$	ε -covering number under KL divergence	–
$\mathcal{F}_s$	Spectral Barron space of order s	23
$\ \cdot\ _{\mathcal{F}_s}$	Spectral Barron norm of order s	23
$\mathcal{T}(\mathbb{R}^d)$	Schwartz function space	22
$\mathcal{T}'(\mathbb{R}^d)$	Space of tempered distributions	22
$\mathbb{P}_s$	Dictionary of ReLU ^s activation functions	16
$\mathcal{H}(\mathbb{D})$	Variation space generated by the dictionary $\mathbb{D}$	16
$\ \cdot\ _{\mathbb{D}}$	Variation norm induced by the dictionary $\mathbb{D}$	16
$\text{KL}(P\ Q)$	Kullback-Leibler divergence	–
$I(X; Y)$	Mutual information	–
$\sim$	Asymptotically equivalent / identically distributed	–
$\hookrightarrow$	Embedding of spaces	–

Appendix B. Proof of Theorem 1

We prove Theorem 1 through the variation space defined by a Minkowski functional (Siegel and Xu, 2024, 2023). Let X be a separable Banach space and let $\mathbb{D} \subset X$ be a uniformly bounded set, often referred to as a *dictionary*. Consider the closed symmetric convex hull of $\mathbb{D}$ in X :

$$\overline{\text{conv}(\pm\mathbb{D})} = \overline{\left\{ \sum_{j=1}^n a_j h_j : n \in \mathbb{N}, h_j \in \mathbb{D}, \sum_{i=1}^n |a_i| \leq 1 \right\}}$$

and the Minkowski functional induced by it:

$$\|f\|_{\mathbb{D}} = \inf \left\{ c > 0 : \frac{f}{c} \in \overline{\text{conv}(\pm\mathbb{D})} \right\},$$

which we call the *variation norm* induced by $\mathbb{D}$. We call

$$\mathcal{K}(\mathbb{D}) = \{f \in X : \|f\|_{\mathbb{D}} < \infty\}$$

the *variation space* generated by $\mathbb{D}$.

Consider the dictionary

$$\mathbb{P}_s = \{\sigma_s(\mathbf{w}^\top \mathbf{x} + b) : \mathbf{w} \in \mathbb{S}^{d-1}, b \in [-1, 1]\} \subset L^2,$$

The set $\mathbb{P}_s$ is uniformly bounded, and as $(\mathbf{w}, b)$ traverses $\mathbb{S}^{d-1} \times [-1, 1]$, the hyperplane $\mathbf{w}^\top \mathbf{x} + b$ sweeps across Ω . Consequently, polynomials of degree at most s on $\mathbf{w}$ can be expressed as linear combinations of $\mathbb{P}_s$ (Siegel and Xu, 2023). Functions in the variation space can be written in integral form, as stated in the following proposition.

Lemma 13 (Yang and Zhou (2025)) *Let $s \in \mathbb{N}_0$. Then*

$$\overline{\text{conv}(\pm\mathbb{P}_s)} = \left\{ f(x) = \int_{\mathbb{S}^{d-1} \times [-1, 1]} \sigma_s(\mathbf{w}^\top \mathbf{x} + b) d\mu(\mathbf{w}, b) : \|\mu\| \leq 1 \right\},$$

where μ is a signed Radon measure on $\mathbb{S}^{d-1} \times [-1, 1]$, and

$$\|f\|_{\mathbb{P}_s} = \inf \left\{ \|\mu\| : f(x) = \int_{\mathbb{S}^{d-1} \times [-1, 1]} \sigma_s(\mathbf{w}^\top \mathbf{x} + b) d\mu(\mathbf{w}, b) \right\}. \quad (1)$$

This integral representation space has been studied in many works (Yang and Zhou, 2025; Chen et al., 2025; Bach, 2017). On the compact set $\Omega = \mathbb{B}^d$ (similar results hold for compact sets satisfying $\text{Span}\{\Omega\} = \mathbb{R}^d$), the variation space $\mathcal{K}(\mathbb{P}_s)$ and the Barron space $\mathcal{B}_s$ are essentially equivalent. Siegel and Xu (2023) proved this for the case $s = 1$; we extend their result to $s \in \mathbb{N}_0$.

Lemma 14 *Let $s \in \mathbb{N}_0$. Then $\mathcal{K}(\mathbb{P}_s) = \mathcal{B}_s$ and*

$$2^{-\frac{d}{2}} \|f\|_{\mathcal{B}_s} \lesssim_s \|f\|_{\mathbb{P}_s} \lesssim_s \|f\|_{\mathcal{B}_s}.$$

Proof We prove only the case $s \geq 1$; the proof for $s = 0$ is entirely similar. Consider the dictionary

$$\mathbb{B} = \{(\|\mathbf{w}\|_1 + |b|)^{-s} \sigma_s(\mathbf{w}^\top \mathbf{x} + b) : \mathbf{w} \in \mathbb{R}^d, b \in \mathbb{R}\} \subset L^2.$$

The dictionary $\mathbb{B}$ is uniformly bounded and compact in L^2 . For any $f \in \mathcal{K}(\mathbb{B})$,

$$f(x) = \int_{\mathbb{B}} (\|\mathbf{w}\|_1 + |b|)^{-s} \sigma_s(\mathbf{w}^\top \mathbf{x} + b) d\mu,$$

where μ is a signed Radon measure on $\mathbb{B}$. Perform a Jordan decomposition $\mu = \mu^+ - \mu^-$ and let $\nu = \mu^+ + \mu^-$. Let $f(\mathbf{w}, b) = \frac{d\mu}{d\nu}(\mathbf{w}, b)$ be the Radon-Nikodym derivative of μ with respect to ν . Define $a = a(\mathbf{w}, b) = (\|\mathbf{w}\|_1 + |b|)^{-s} f(\mathbf{w}, b) \|\nu\|$ and make the change of variable $\rho = \frac{\nu}{\|\nu\|}$ to obtain

$$f(x) = \mathbb{E}_{(a, \mathbf{w}, b) \sim \rho} [a \sigma_s(\mathbf{w}^\top \mathbf{x} + b)], \quad x \in \Omega$$

and

$$\|\mu\| = \mathbb{E}_{(a, \mathbf{w}, b) \sim \rho} [a |(\|\mathbf{w}\|_1 + |b|)^s].$$

Conversely, for any $f \in \mathcal{B}_s$, one can also find a measure μ on $\mathbb{B}$ such that $f = \int_{\mathbb{B}} i_{\mathbb{B} \rightarrow L^2} d\mu$ and the above equation holds. Thus, $\mathcal{K}(\mathbb{B}) = \mathcal{B}_s$ and $\|f\|_{\mathcal{K}(\mathbb{B})} = \|f\|_{\mathcal{B}_s}$.

We prove the lower bound first. It is enough to show

$$\mathbb{P}_s \subset C \overline{\text{conv}(\pm \mathbb{B})}.$$

Indeed, for any $g(x) = \sigma_s(\mathbf{w}^\top \mathbf{x} + b) \in \mathbb{P}_s$, where $\mathbf{w} \in \mathbb{S}^{d-1}, b \in I$, since

$$(\|\mathbf{w}\|_1 + |b|)^s \leq 2^s,$$

and $(\|\mathbf{w}\|_1 + |b|)^{-s} \sigma_s(\mathbf{w}^\top \mathbf{x} + b) \in \mathbb{B}$, it follows that $g \in Cd^{\frac{s}{2}} \overline{\text{conv}(\pm \mathbb{B})}$. Consequently, $\overline{\text{conv}(\pm \mathbb{P}_s)} \subset Cd^{\frac{s}{2}} \overline{\text{conv}(\pm \mathbb{B})}$. For any $c > 0$ satisfying $\frac{f}{c} \in \overline{\text{conv}(\pm \mathbb{P}_s)}$, we have $f \in cCd^{\frac{s}{2}} \overline{\text{conv}(\pm \mathbb{B})}$, and thus

$$\|f\|_{\mathcal{B}_s} = \|f\|_{\mathbb{B}} \leq cCd^{\frac{s}{2}}.$$

By the arbitrariness of c , the first inequality follows.

For the upper bound, a similar argument reduces the proof to showing $\mathbb{B} \subset \overline{\text{conv}(\pm \mathbb{P}_s)}$. For any $g = (\|\mathbf{w}\|_1 + |b|)^{-s} \sigma_s(\mathbf{w}^\top \mathbf{x} + b) \in \mathbb{B}$, where $\mathbf{w} \in \mathbb{R}^d, b \in \mathbb{R}$. If $\|\mathbf{w}\|_1 \neq 0$, by the positive homogeneity of degree s of σ_s , we may assume without loss of generality that $\mathbf{w} \in \mathbb{S}^{d-1}$. Since for $\mathbf{x} \in \Omega = \mathbb{B}^d$, we have $(\|\mathbf{w}\|_1 + |b|)^{-s} \leq (1 + |b|)^{-s}$, it suffices to prove

$$\tilde{g}(x) = (1 + |b|)^{-s} \sigma_s(\mathbf{w}^\top \mathbf{x} + b) \in C \overline{\text{conv}(\pm \mathbb{P}_s)}.$$

If $b \in I$, then taking $C = 1$ suffices. If $b < -1$, then $\tilde{g} \equiv 0 \in \overline{\text{conv}(\pm \mathbb{P}_s)}$. If $b > 1$, then

$$\tilde{g}(x) = (1 + |b|)^{-s} (\mathbf{w}^\top \mathbf{x} + b)^s = (1 + |b|)^{-s} \sum_{i=0}^s \binom{s}{i} b^{s-i} (\mathbf{w}^\top \mathbf{x})^i.$$

By (Bach, 2017, discussion after Proposition 6), there exists an absolute constant C such that for $i = 0, \dots, s$, $(\mathbf{w}^\top \mathbf{x})^i \in C \overline{\text{conv}(\pm \mathbb{P}_s)}$. Moreover,

$$(1 + |b|)^{-s} \sum_{i=1}^s \binom{s}{i} |b|^{s-i} = \left(\frac{1 + |b|}{1 + |b|} \right)^s \leq 1,$$

hence $\tilde{g} \in c(d, s) \overline{\text{conv}(\pm \mathbb{P}_s)}$. If $\|\mathbf{w}\|_1 = 0$, then $g(x)$ is identically 1 or 0 on Ω , and thus also belongs to $C \overline{\text{conv}(\pm \mathbb{P}_s)}$. This completes the proof of Lemma 14. $\blacksquare$

Proof of Theorem 1. This is a direct corollary of (Siegel, 2025, Theorem 3) and Lemma 14. This completes the proof of Theorem 1. $\blacksquare$

Appendix C. Proof of Theorem 2

This appendix follows the approach of (Bach, 2017; Yang and Zhou, 2025) to prove Theorem 2. We first transform the problem into an approximation problem on the sphere. To use spherical harmonic analysis, we take the parameter space in this section to be $\mathbb{S}^d$, instead of $\mathbb{S}^{d-1} \times I$, and write the parameter as $\boldsymbol{\theta} = (\mathbf{w}^\top, b) \in \mathbb{S}^d$. We begin with a brief review of harmonic analysis on the sphere (Dai and Xu, 2013). For $i \in \mathbb{N}_0$, the space of spherical harmonics of degree i , $\mathbb{Y}_i$, is the linear space of restrictions to $\mathbb{S}^d$ of real homogeneous harmonic polynomials of degree i on $\mathbb{R}^{d+1}$. The dimension of $\mathbb{Y}_i$ is

$$N(d, i) := \begin{cases} \frac{2i+d-1}{i} \binom{i+d-2}{d-1}, & i \neq 0 \\ 1, & i = 0. \end{cases}$$

Spherical harmonics are eigenfunctions of the Laplace-Beltrami operator Δ :

$$\Delta Y_i = -i(i+d-1)Y_i, \quad Y_i \in \mathbb{Y}_i, \quad (2)$$

where, in coordinates $\mathbf{u} = (u_1, \dots, u_{d+1}) \in \mathbb{S}^d$,

$$\Delta = \sum_{k=1}^d \frac{\partial^2}{\partial u_k^2} - \sum_{k=1}^d \sum_{\ell=1}^d u_k u_\ell \frac{\partial^2}{\partial u_k \partial u_\ell} - d \sum_{k=1}^d u_k \frac{\partial}{\partial u_k}.$$

Define the inner product $\langle g_1, g_2 \rangle = \int_{\mathbb{S}^d} g_1(\mathbf{u}) g_2(\mathbf{u}) d\tau_d(\mathbf{u})$ on $\mathbb{S}^d$; then spherical harmonics of different degrees are orthogonal with respect to this inner product.

Any $g \in L^2(\mathbb{S}^d)$ has a unique decomposition in $L^2(\mathbb{S}^d)$:

$$g(\mathbf{u}) = \int_{\mathbb{S}^d} g(\mathbf{v}) d\tau_d(\mathbf{v}) + \sum_{i=1}^{\infty} N(d, i) \int_{\mathbb{S}^d} g(\mathbf{v}) P_i(\mathbf{u}^\top \mathbf{v}) d\tau_d(\mathbf{v}) =: \sum_{i=0}^{\infty} g_i(\mathbf{u}),$$

where P_i are the Gegenbauer polynomials:

$$P_i(t) := \frac{(-1)^i}{2^i} \frac{\Gamma(d/2)}{\Gamma(i+d/2)} (1-t^2)^{(2-d)/2} \left(\frac{d}{dt} \right)^i (1-t^2)^{i+(d-2)/2}, \quad t \in [-1, 1],$$

which satisfy $\|P_i\|_{L^\infty[-1,1]} \leq 1$ and $1 - P_i(\cos \beta) \asymp \min\{1, i^2 \beta^2\}$. The above decomposition can be viewed as an extension of Fourier series to the sphere. For $g \in L^2(\mathbb{S}^d)$, denote by g_i its projection onto the spherical harmonic space of degree i . We also have the Parseval identity on the sphere:

$$\|g\|_{L^2(\mathbb{S}^d)}^2 = \sum_{i=0}^{\infty} \|g_i\|_{L^2(\mathbb{S}^d)}^2. \quad (3)$$

A useful observation is that if $g \in \tilde{\mathcal{F}}_{p, \tau_d, s}$ and $g(\mathbf{u}) = \int_{\mathbb{S}^d} a(\boldsymbol{\theta}) \sigma_s(\boldsymbol{\theta}^\top \mathbf{u}) d\tau_d(\boldsymbol{\theta})$, then the Funk-Hecke formula (Bach, 2017) gives the Fourier series decomposition

$$g(\mathbf{u}) = \sum_{i=0}^{\infty} \lambda_i a_i(\mathbf{u}), \quad (4)$$

where λ_i denotes the Fourier coefficient of g with respect to the Gegenbauer polynomial:

$$\lambda_i := \frac{\omega_{d-1}}{\omega_d} \int_{-1}^1 g(t) P_i(t) (1-t^2)^{\frac{d-2}{2}} dt.$$

Bach (2017) provided specific calculations and estimates for these coefficients:

$$\lambda_i = \begin{cases} \frac{\omega_{d-1}}{\omega_d} \frac{\Gamma(\frac{d}{2})\Gamma(\frac{s+1}{2})}{2\Gamma(\frac{s+d+1}{2})}, & i = 0 \\ 0, & i \equiv s \pmod{2}, i \geq s+1 \\ \frac{\Gamma(\frac{d}{2})\Gamma(i-s)}{\Gamma(\frac{i-s+1}{2})\Gamma(\frac{i+d+s+1}{2})}, & i \equiv s+1 \pmod{2}, i \geq s+1 \end{cases}.$$

By Stirling's formula, when $\lambda_i \neq 0$, we have the asymptotic estimate

$$\lambda_i \asymp i^{-\frac{d+2s+1}{2}}. \quad (5)$$

In particular, when $p = 2$, by the Parseval identity (3), we can express the $\tilde{\mathcal{F}}_{2,\tau_d,s}$ norm of g as

$$\|g\|_{\tilde{\mathcal{F}}_{2,\tau_d,s}}^2 = \sum_{\lambda_i \neq 0} \frac{1}{\lambda_i^2} \|g_i\|_{L^2(\mathbb{S}^d)}^2.$$

It should be noted that for a general measure $\nu \neq \tau_d$, functions in $\tilde{\mathcal{F}}_{p,\nu,s}(\mathbb{S}^d)$ generally do not admit such a simple Fourier series decomposition and norm expression; therefore, we choose $\nu = \tau_d$ in our analysis. For approximation theorems under more general probability measures, more refined analytical tools may be required.

Next, we introduce the smoothness of functions on the sphere. For $0 \leq \beta \leq \pi$, the translation operator T_β (spherical mean operator) is defined as

$$T_\beta g(\mathbf{u}) := \int_{\mathbb{S}_u^\perp} g(\mathbf{u} \cos \beta + \mathbf{v} \sin \beta) d\tau_{d-1}(\mathbf{v}), \quad \mathbf{u} \in \mathbb{S}^d, g \in L^1(\mathbb{S}^d),$$

where $\mathbb{S}_u^\perp := \{\mathbf{v} \in \mathbb{S}^d : \mathbf{u}^\top \mathbf{v} = 0\}$ is the equator of $\mathbb{S}^d$ relative to $\mathbf{u}$ (isomorphic to $\mathbb{S}^{d-1}$). For $\alpha > 0$ and $0 < \beta < \pi$, we define the α -th order difference operator

$$\Delta_\beta^\alpha := (I - T_\beta)^{\alpha/2} = \sum_{j=0}^{\infty} (-1)^j \binom{\frac{\alpha}{2}}{j} T_\beta^j,$$

where $\binom{\alpha}{j} = \frac{\alpha(\alpha-1)\cdots(\alpha-j+1)}{j!}$, and it satisfies

$$(\Delta_\beta^\alpha g)_i = (1 - P_i(\cos \beta))^{\frac{\alpha}{2}} g_i, \quad i \in \mathbb{N}_0. \quad (6)$$

For $g \in L^p(\mathbb{S}^d)$ with $1 \leq p \leq 2$, its α -th order modulus of smoothness is defined as

$$\omega_\alpha(g, t)_p := \sup_{0 < \beta \leq t} \|\Delta_\beta^\alpha g\|_{L^p(\mathbb{S}^d)}, \quad 0 < t < \pi.$$

With the above preparations, we now prove the intermediate approximation results stated in the main text.

Proof of Lemma 3. By homogeneity, assume $\|\tilde{f}\|_{\tilde{\mathcal{F}}_{p,\tau_d,s}(\mathbb{S}^d)} \leq 1$. Write

$$\tilde{f}(\mathbf{u}) = \int_{\mathbb{S}^d} a(\boldsymbol{\theta}) \sigma_s(\boldsymbol{\theta}^\top \mathbf{u}) d\tau_d(\boldsymbol{\theta}), \quad \|a\|_{L^p(\mathbb{S}^d)} \leq 1.$$

We first estimate the modulus of smoothness of $\tilde{f}$. From the previous discussion, the multipliers λ_i in the spherical harmonic expansion of the activation function σ_s satisfy: when $i \geq s+1$ and

$i + 1 \equiv s \pmod{2}$, $\lambda_i \asymp i^{-\frac{d+2s+1}{2}}$; otherwise $\lambda_i = 0$. Denote the set of such indices by $\mathcal{I}_s$. Write $\tilde{f} = \sum_{i \in \mathcal{I}_s} \lambda_i a_i$. By the Parseval identity (3) and (6), we have

$$\|\Delta_\beta^\alpha \tilde{f}\|_{L^2(\mathbb{S}^d)}^2 = \sum_{i \in \mathcal{I}_s} \lambda_i^2 (1 - P_i(\cos \beta))^\alpha \|a_i\|_{L^2(\mathbb{S}^d)}^2.$$

According to the spherical harmonic projection theorem (Dai and Xu, 2013, Theorem 9.1.1),

$$\|a_i\|_{L^2(\mathbb{S}^d)} \lesssim i^{\frac{Q(p,d)}{2p}},$$

where

$$Q(p, d) = \begin{cases} 2d - pd - p, & 1 \leq p \leq \frac{2d+2}{d+3}, \\ \frac{1}{2}(pd - 2d + p - 2), & \frac{2d+2}{d+3} < p \leq 2. \end{cases}$$

Set $\gamma := \frac{Q(p,d)}{p} - (d+2s+1)$. Substituting the asymptotic estimate for λ_i from (5) and $1 - P_i(\cos \beta) \asymp \min(1, i^{2\alpha} \beta^{2\alpha})$, we obtain

$$\|\Delta_\beta^\alpha \tilde{f}\|_{L^2}^2 \lesssim \sum_{i \in \mathcal{I}_s} i^\gamma \min(1, i^{2\alpha} \beta^{2\alpha}) \lesssim \sum_{k=1}^{\infty} k^\gamma \min(1, k^{2\alpha} \beta^{2\alpha}).$$

Let $K_\beta = \left\lfloor \frac{1}{\beta} \right\rfloor$ and split the sum into two parts: $k \leq K_\beta$ and $k > K_\beta$:

$$S_1 = \beta^{2\alpha} \sum_{k=1}^{K_\beta} k^{\gamma+2\alpha}, \quad S_2 = \sum_{k=K_\beta+1}^{\infty} k^\gamma.$$

Note that by the definition of $Q(p, d)$, we always have $\gamma < -1$, so the series S_2 always converges. If $\gamma + 2\alpha > -1$, then $S_1 \asymp \beta^{2\alpha} K_\beta^{\gamma+2\alpha+1} \asymp \beta^{-\gamma-1}$, and $S_2 \asymp K_\beta^{\gamma+1} \asymp \beta^{-\gamma-1}$. If $\gamma + 2\alpha = -1$, then $S_1 \asymp \beta^{2\alpha} \log K_\beta \asymp \beta^{-\gamma-1} \log \frac{1}{\beta}$, and $S_2 \asymp \beta^{-\gamma-1}$. If $\gamma + 2\alpha < -1$, then $S_1 \asymp \beta^{2\alpha}$, and $S_2 \asymp \beta^{-\gamma-1}$. In this case, since $\gamma < -1$ and $2\alpha < -\gamma - 1$, the term $\beta^{2\alpha}$ dominates. Combining the above, we get

$$\|\Delta_\beta^\alpha \tilde{f}\|_{L^2}^2 \lesssim \begin{cases} \beta^{2\alpha}, & \gamma + 2\alpha < -1, \\ \beta^{-\gamma-1} \log \frac{1}{\beta}, & \gamma + 2\alpha = -1, \\ \beta^{-\gamma-1}, & \gamma + 2\alpha > -1. \end{cases}$$

For $0 < t \leq \pi$, since $\omega_\alpha(\tilde{f}, t)_2 = \sup_{0 < \beta \leq t} \|\Delta_\beta^\alpha \tilde{f}\|_{L^2}$ and the estimate is monotonic, we have

$$\omega_\alpha(\tilde{f}, t)_2 \lesssim \begin{cases} t^\alpha, & \alpha < \gamma^* \\ t^{\gamma^*} \sqrt{\log \frac{1}{t}}, & \alpha = \gamma^*, \\ t^{\gamma^*}, & \alpha > \gamma^* \end{cases}, \quad (7)$$

where $\gamma^* = -\frac{\gamma+1}{2}$ is the critical value.

Consider $g_j := \int_{\mathbb{S}^d} \tilde{f}(\mathbf{u}) L_j(\mathbf{u}^\top \mathbf{v}) d\tau_d(\mathbf{v})$, where

$$L_j(t) := \sum_{i=0}^{\infty} \eta\left(\frac{i}{j}\right) N(d, i) P_i(t), \quad j \in \mathbb{N}_0,$$

Here η is a C^∞ function on $[0, \infty)$ such that $\eta(t) = 1$ for $0 \leq t \leq 1$ and $\eta(t) = 0$ for $t \geq 2$. We claim that $g_j \in \tilde{\mathcal{F}}_{2, \tau_d, s}(\mathbb{S}^d)$ and that for sufficiently large j , g_j approximates $\tilde{f}$. Since η is supported

on $[0, 2]$, the sum can be terminated at $i = 2j - 1$, so g_j is a polynomial of degree at most $2j - 1$. Because $\tilde{f}$ is an odd (even) function when k is even (odd), we have $(g_j)_i = \eta(i/j)\tilde{f}_i = 0$ for any $i \equiv k \pmod{2}$. Moreover, (Dai and Xu, 2013, Theorem 10.3.2) shows that

$$\omega_\alpha(\tilde{f}, j^{-1})_2 \asymp \|\tilde{f} - g_j\|_{L^2(\mathbb{S}^d)} + j^{-\alpha} \|(-\Delta)^{\alpha/2} g_j\|_{L^2(\mathbb{S}^d)}.$$

Substituting the estimate (7) into the above, we obtain

$$\|(-\Delta)^{\alpha/2} g_j\|_{L^2(\mathbb{S}^d)} \lesssim \begin{cases} 1, & \alpha < \gamma^* \\ \sqrt{\log j}, & \alpha = \gamma^* \\ j^{\alpha - \gamma^*}, & \alpha > \gamma^* \end{cases}.$$

Set $\alpha^* = s + \frac{d+1}{2}$. By (2), $((-\Delta)^{\alpha^*/2} g_j)_i = (i(i+d-1))^{\alpha^*/2} (g_j)_i$. We can estimate the norm $\|g_j\|_{\tilde{\mathcal{F}}_{2, \tau_d, s}}$ as follows:

$$\begin{aligned} \|g_j\|_{\tilde{\mathcal{F}}_{2, \tau_d, s}}^2 &= \sum_{\lambda_i \neq 0} \frac{1}{\lambda_i^2} \|(g_j)_i\|_{L^2(\mathbb{S}^d)}^2 \\ &\lesssim \lambda_0 \|(g_j)_0\|_{L^2(\mathbb{S}^d)}^2 + \sum_{i=1}^{2j-1} i^{d+2s+1} i^{-2\alpha^*} \|((-\Delta)^{\alpha^*/2} g_j)_i\|_{L^2(\mathbb{S}^d)}^2 \\ &\lesssim 1 + \sum_{i=1}^{2j-1} i^{d+2s+1-2\alpha^*} \|((-\Delta)^{\alpha^*/2} g_j)_i\|_{L^2(\mathbb{S}^d)}^2 \\ &\lesssim 1 + \|(-\Delta)^{\alpha^*/2} g_j\|_{L^2(\mathbb{S}^d)}^2, \end{aligned}$$

where the last inequality follows from the choice of α^* and the Parseval identity (3).

- If $1 \leq p < \frac{2d+2}{d+3}$, then $\alpha^* > \gamma^*$, and thus

$$\|g_j\|_{\tilde{\mathcal{F}}_{2, \tau_d, s}}^2 \lesssim j^{2\alpha^* - 2\gamma^*} = j^{\frac{2d}{p} - d}.$$

Choose $j \asymp R^{\frac{2p}{d(2-p)}}$. Then $\|g_j\|_{\tilde{\mathcal{F}}_{2, \tau_d, s}} \lesssim R$, and

$$\|\tilde{f} - g_j\|_{L^2(\mathbb{S}^d)} \lesssim j^{-\alpha^*} = R^{-\frac{2s+d+1}{d} \cdot \frac{p}{2-p}}.$$

- If $p = \frac{2d+2}{d+3}$, then $\alpha^* = \gamma^*$, and thus

$$\|g_j\|_{\tilde{\mathcal{F}}_{2, \tau_d, s}}^2 \lesssim \log j.$$

Choose $j \asymp e^{R^2}$. Then $\|g_j\|_{\tilde{\mathcal{F}}_{2, \tau_d, s}} \lesssim R$, and

$$\|\tilde{f} - g_j\|_{L^2(\mathbb{S}^d)} \lesssim j^{-\alpha^*} \sqrt{\log j} = R e^{-(s + \frac{d+1}{2}) R^2}.$$

- If $\frac{2d+2}{d+3} < p \leq 2$, then $\alpha^* > \gamma^*$, and thus

$$\|g_j\|_{\tilde{\mathcal{F}}_{2, \tau_d, s}}^2 \lesssim j^{2\alpha^* - 2\gamma^*} = j^{\frac{d+3}{2} - \frac{d+1}{p}}.$$

Choose $j \asymp R^{\frac{4p}{p(d+3)-2(d+1)}}$. Then $\|g_j\|_{\tilde{\mathcal{F}}_{2, \tau_d, s}} \lesssim R$, and

$$\|\tilde{f} - g_j\|_{L^2(\mathbb{S}^d)} \lesssim j^{-\alpha^*} = R^{-\frac{2p(2s+d+1)}{p(d+3)-2(d+1)}}.$$

This completes the proof of Lemma 3. ■

Lemma 3 approximates functions in $\tilde{\mathcal{F}}_{p,\tau_d,s}(\mathbb{S}^d)$ by functions in $\tilde{\mathcal{F}}_{2,\tau_d,s}(\mathbb{S}^d)$. The key point is to use the boundedness theorem for spherical-harmonic projection operators to convert the L^p integrability of the weight a into spectral decay, thereby estimating the modulus of smoothness. We next prove the lifting step from the sphere to $\mathbb{B}^d$.

Proof of Corollary 4. Denote $\mathbb{S}_{Cap}^d = \left\{ (\mathbf{u}_1, \dots, u_{d+1})^\top \in \mathbb{S}^d : u_{d+1} \geq \frac{\sqrt{2}}{2} \right\}$. For any $f \in \tilde{\mathcal{F}}_{p,\tau_d,s}$, first lift it to the sphere $\mathbb{S}^d$. Specifically, define

$$\tilde{f}(\mathbf{u}) := u_{d+1}^s f\left(\frac{\mathbf{u}'}{u_{d+1}}\right), \quad \mathbf{u} = (\mathbf{u}_1, \dots, u_{d+1})^\top \in \mathbb{S}_{Cap}^d,$$

where $\mathbf{u}' = (u_1, \dots, u_d)^\top$. Using the Whitney extension theorem (Fefferman, 2006), we can extend $\tilde{f}$ to the entire sphere $\mathbb{S}^d$ (still denoted $\tilde{f}$) while preserving smoothness, so that $\tilde{f} \in \tilde{\mathcal{F}}_{p,\tau_d,s}(\mathbb{S}^d)$. By multiplying $\tilde{f}$ by a smooth cutoff function (equal to 1 for $u_{d+1} > \frac{\sqrt{2}}{2}$ and 0 for $u_{d+1} \leq 0$) and performing odd/even extension on the lower hemisphere $u_{d+1} < 0$, we can choose the extended $\tilde{f}$ to be an odd or even function.

Conversely, for $g \in \tilde{\mathcal{F}}_{2,\tau_d,s}(\mathbb{S}^d)$, define the reverse mapping

$$g^*(\mathbf{x}) := (\|\mathbf{x}\|_2^2 + 1)^{\frac{s}{2}} g\left(\frac{1}{\sqrt{\|\mathbf{x}\|_2^2 + 1}} \begin{pmatrix} \mathbf{x} \\ 1 \end{pmatrix}\right), \quad \mathbf{x} \in \mathbb{B}^d.$$

One can verify that $g^* \in \tilde{\mathcal{F}}_{2,\tau_d,s}(\Omega)$, and the error estimate is

$$\|f - g^*\|_{L^2(\Omega)} = \|\tilde{f}^* - g^*\|_{L^2(\Omega)} \leq \max\{1, 2^{s/2}\} \|\tilde{f} - g\|_{L^2(\mathbb{S}^d)},$$

which, combined with Lemma 3, yields the conclusion. This completes the proof of Corollary 4. ■

Proof of Theorem 2. Take the case $1 \leq p < p^*$ as an example; other values of p are similar. Without loss of generality, assume $\|f\|_{\tilde{\mathcal{F}}_{p,\tau_d,s}} = 1$. By Corollary 4, there exists $g \in \tilde{\mathcal{F}}_{2,\tau_d,s}$ with $\|g\|_{\tilde{\mathcal{F}}_{2,\tau_d,s}} \leq R$ such that

$$\|f - g\|_{L^2(\Omega)} \lesssim R^{-\frac{2s+d+1}{d} \cdot \frac{p}{2-p}},$$

and by (Chen et al., 2025, Theorem 6.3), there exists a random-feature model (a special shallow ReLU^s network) f_m such that

$$\|g - f_m\|_{L^2(\Omega)} \lesssim R m^{-\frac{1}{2}} \log^{\frac{3}{2}} m,$$

hence

$$\|f - f_m\|_{L^2(\Omega)} \leq \|f - g\|_{L^2(\Omega)} + \|g - f_m\|_{L^2(\Omega)} \lesssim R^{-\frac{2s+d+1}{d} \cdot \frac{p}{2-p}} + R m^{-\frac{1}{2}} \log^{\frac{3}{2}} m,$$

Choosing $R \gtrsim m^{\frac{d(2-p)}{2d(2-p)+2p(2s+d+1)}}$ yields the upper bound $O\left(m^{-\frac{1}{2} - \frac{d(2-p)}{2d(2-p)+2p(2s+d+1)}} \log^{\frac{3}{2}} m\right)$. This completes the proof of Theorem 2. ■

Appendix D. Proof of Theorem 5

We prove Theorem 5 by combining an embedding into spectral Barron spaces with the corresponding shallow-network approximation result. Let $\mathcal{S}(\mathbb{R}^d)$ and $\mathcal{S}'(\mathbb{R}^d)$ denote the Schwartz space and the

space of tempered distributions on $\mathbb{R}^d$, respectively, with the Fourier transform understood in the distributional sense. For $s \geq 0$, define the *spectral Barron space of order s* as follows:

$$\mathcal{F}_s = \left\{ f(\mathbf{x}) = \int_{\mathbb{R}^d} e^{i\mathbf{x} \cdot \xi} \hat{F}(d\xi) : \hat{F} \text{ is a finite complex measure, } \|f\|_{\mathcal{F}_s(\mathbf{w})} < \infty \right\}.$$

Here,

$$\|f\|_{\mathcal{F}_s} = \inf_{f_e|_{\mathbf{w}=f}} \int_{\mathbb{R}^d} (1 + \|\xi\|_1)^s |\hat{F}_e(d\xi)|,$$

where the infimum is taken over all $\mathcal{L}'(\mathbb{R}^d)$ extensions of f . For spectral Barron-type spaces, Xu (2020) proved that for $s \in \mathbb{N}$ and $f \in \mathcal{F}^s$,

$$\inf_{f_m \in \Sigma_m^{s-1}} \|f - f_m\|_{L^2} \lesssim \|f\|_{\mathcal{F}_s} m^{-\frac{1}{2} - \frac{1}{d}},$$

and the uniform approximation error (see (Ma et al., 2022, Theorem 5)):

$$\inf_{f_m \in \Sigma_m^k} \|f - f_m\|_{L^\infty} \lesssim \|f\|_{\mathcal{F}_s} m^{-\frac{1}{2} - \frac{s-1}{d+1}} \begin{cases} \log m, & 1 < s < k(d+1) + 1 \\ (\log m)^{1+k}, & s = k(d+1) + 1 \end{cases}, \quad (8)$$

Furthermore, Liao and Ming (2025) and Choulli et al. (2025) established the embedding relationship between Sobolev spaces and spectral Barron spaces: for $s \in \mathbb{N}_0$, $1 \leq p \leq 2$, if $\alpha > s + \frac{d}{p}$, then

$$W^{\alpha,p} \hookrightarrow \mathcal{F}_s.$$

For the critical case $p = 1$ and $\alpha = s + d$, the above embedding also holds, i.e.,

$$W^{s+d,1} \hookrightarrow \mathcal{F}_s.$$

We use this embedding to prove the approximation theorem for Sobolev spaces, Theorem 5.

Proof of Theorem 5. We only need to prove the case $1 \leq p < 2$ and $\alpha < s + d$. Assume Ω is contained in a ball of radius $R > 0$, i.e., $\mathbb{B}_R^d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 \leq R\}$. By the classical Sobolev extension theorem, f can be extended to $\mathbb{R}^d$ with $\|f\|_{W^{\alpha,p}(\mathbb{R}^d)} \lesssim_\Omega \|f\|_{W^{\alpha,p}}$. Moreover, we can require that f is supported in $\mathbb{B}_{R+1}^d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 \leq R+1\}$.

Let $\phi : \mathbb{R}^d \rightarrow [0, +\infty)$ be a C^∞ radially symmetric function supported in $\mathbb{B}_R^d$, satisfying $\int_{\mathbb{R}^d} \phi(\mathbf{x}) d\mathbf{x} = 1$. For any $\varepsilon > 0$, define $\phi_\varepsilon(\mathbf{x}) = \varepsilon^{-d} \phi(\mathbf{x}/\varepsilon)$. Then ϕ_ε is an approximate identity, i.e., $\lim_{\delta \rightarrow 0} \|\phi_\delta * f - f\|_{L^p} = 0$. By the triangle inequality and the normalization of ϕ , for any $\delta > 0$, we have

$$\|\phi_\delta * f\|_{W^{\alpha,p}(\mathbb{R}^d)} \leq \|f\|_{W^{\alpha,p}(\mathbb{R}^d)}.$$

Therefore, we may assume $f \in C_c^\infty(\mathbb{R}^d)$ without loss of generality.

Now let $\varepsilon > 0$ be a parameter to be determined. Define the approximant

$$f_\varepsilon(\mathbf{x}) = \sum_{t=1}^{\alpha} \binom{\alpha}{t} (-1)^{t-1} \int_{\mathbb{R}^d} \phi_\varepsilon(\mathbf{y}) f(\mathbf{x} - t\mathbf{y}) d\mathbf{y}. \quad (9)$$

Using the normalization of ϕ_ε , we estimate $\|f - f_\varepsilon\|_{L^p(\mathbb{R}^d)}$:

$$\|f - f_\varepsilon\|_{L^p(\mathbb{R}^d)} \leq \left\| \int_{\mathbb{R}^d} \phi_\varepsilon(\mathbf{y}) \sum_{t=0}^{\alpha} \binom{\alpha}{t} (-1)^t f(\mathbf{x} - t\mathbf{y}) d\mathbf{y} \right\|_{L^p(\mathbb{R}^d)}.$$

For a fixed $\mathbf{y} \in \mathbb{R}^d$, we estimate

$$\left\| \left(\sum_{t=0}^{\alpha} \binom{\alpha}{t} (-1)^t f(\mathbf{x} - t\mathbf{y}) \right) \right\|_{L^p(\mathbb{R}^d)} = \|\Delta_{\mathbf{y}}^{\alpha} f\|_{L^p(\mathbb{R}^d)},$$

where $\Delta_{\mathbf{y}}^{\alpha}$ is the α -th order difference operator of f . By the fundamental theorem of calculus,

$$\Delta_{\mathbf{y}} f(\mathbf{x}) = - \int_0^1 \nabla f(\mathbf{x} - t\mathbf{y}) \cdot \mathbf{y} dt.$$

Since the difference operator $\Delta_{\mathbf{y}}$ commutes with integration over t , we have

$$\Delta_{\mathbf{y}}^{\alpha} f(\mathbf{x}) = (-1)^{\alpha} \int_{[0,1]^{\alpha}} D^{\alpha} f(\mathbf{x} - (t_1 + \dots + t_{\alpha})\mathbf{y}) \cdot \mathbf{y}^{\otimes \alpha} dt_1 \dots dt_{\alpha}.$$

For a multi-index $|\gamma| = \alpha$, by properties of mixed partial derivatives, there exists a constant C_{α} such that

$$|\Delta_{\mathbf{y}}^{\alpha} f(\mathbf{x})| \leq C_{\alpha} |\mathbf{y}|^{\alpha} \int_{[0,1]^{\alpha}} |D^{\alpha} f(\mathbf{x} - (t_1 + \dots + t_{\alpha})\mathbf{y})| dt_1 \dots dt_{\alpha},$$

where $C_{\alpha} = \alpha^{\alpha} \binom{\alpha+d-1}{d-1}$ is an upper bound for the polynomial coefficients. If $p = \infty$, then $\|\Delta_{\mathbf{y}}^{\alpha} f\|_{L^{\infty}(\mathbb{R}^d)} \leq C_{\alpha} |\mathbf{y}|^{\alpha} \|f\|_{W^{\alpha, \infty}(\mathbb{R}^d)}$. If $1 \leq p < \infty$, by Hölder's inequality (applied to $(t_1, \dots, t_{\alpha})$ with conjugate exponent $p' = \frac{p}{p-1}$), we have

$$\begin{aligned} |\Delta_{\mathbf{y}}^{\alpha} f(\mathbf{x})|^p &\leq C_{\alpha}^p |\mathbf{y}|^{\alpha p} \left(\int_{[0,1]^{\alpha}} 1^{p'} dt_1 \dots dt_{\alpha} \right)^{\frac{p}{p'}} \int_{[0,1]^{\alpha}} |D^{\alpha} f(\mathbf{x} - (t_1 + \dots + t_{\alpha})\mathbf{y})|^p dt_1 \dots dt_{\alpha} \\ &= C_{\alpha}^p |\mathbf{y}|^{\alpha p} \int_{[0,1]^{\alpha}} |D^{\alpha} f(\mathbf{x} - (t_1 + \dots + t_{\alpha})\mathbf{y})|^p dt_1 \dots dt_{\alpha}, \end{aligned}$$

where we used $\int_{[0,1]^{\alpha}} 1 dt_1 \dots dt_{\alpha} = 1$. Integrating over $\mathbf{x}$ and swapping the order of integration yields

$$\begin{aligned} \|\Delta_{\mathbf{y}}^{\alpha} f\|_{L^p(\mathbb{R}^d)}^p &\leq C_{\alpha}^p |\mathbf{y}|^{\alpha p} \int_{[0,1]^{\alpha}} \int_{\mathbb{R}^d} |D^{\alpha} f(\mathbf{x} - (t_1 + \dots + t_{\alpha})\mathbf{y})|^p d\mathbf{x} dt_1 \dots dt_{\alpha} \\ &= C_{\alpha}^p |\mathbf{y}|^{\alpha p} \|f\|_{W^{\alpha, p}(\mathbb{R}^d)}^p. \end{aligned}$$

Therefore,

$$\|\Delta_{\mathbf{y}}^{\alpha} f\|_{L^p(\mathbb{R}^d)} \leq C_{\alpha} |\mathbf{y}|^{\alpha} \|f\|_{W^{\alpha, p}(\mathbb{R}^d)}.$$

Since ϕ_{ε} is supported in $B_{R\varepsilon}$, we have

$$\begin{aligned} \|f - f_{\varepsilon}\|_{L^p(\mathbb{R}^d)} &\leq \int_{\mathbb{R}^d} \phi_{\varepsilon}(\mathbf{y}) \|\Delta_{\mathbf{y}}^{\alpha} f\|_{L^p(\mathbb{R}^d)} d\mathbf{y} \\ &\leq C_{\alpha} \|f\|_{W^{\alpha, p}(\mathbb{R}^d)} \int_{B_{R\varepsilon}} \phi_{\varepsilon}(\mathbf{y}) |\mathbf{y}|^{\alpha} d\mathbf{y} \\ &\leq C_{\alpha} R^{\alpha} \|f\|_{W^{\alpha, p}(\mathbb{R}^d)} \varepsilon^{\alpha}, \end{aligned}$$

where the last step uses the fact that ϕ_{ε} is supported in $B_{R\varepsilon}$ and $\int \phi_{\varepsilon} = 1$.

Next, we estimate $\|f_\varepsilon\|_{W^{\alpha_0,1}(\mathbb{R}^d)}$, where $\alpha_0 = s + d$. Since α is fixed, it suffices to estimate

$$\left\| \int_{\mathbb{R}^d} \phi_\varepsilon(\mathbf{y}) f(\mathbf{x} - t\mathbf{y}) d\mathbf{y} \right\|_{W^{\alpha_0,1}(\mathbb{R}^d)}, \quad t = 1, \dots, \alpha.$$

To this end, perform a change of variables:

$$f_{\varepsilon,t}(\mathbf{x}) = \frac{1}{t^d} \int_{\mathbb{R}^d} \phi_\varepsilon\left(\frac{\mathbf{y}}{t}\right) f(\mathbf{x} - \mathbf{y}) d\mathbf{y} = \int_{\mathbb{R}^d} \phi_{t\varepsilon}(\mathbf{y}) f(\mathbf{x} - \mathbf{y}) d\mathbf{y}.$$

Consider

$$\|f_{\varepsilon,t}\|_{W^{\alpha_0,1}(\mathbb{R}^d)} = \sum_{|\beta| \leq \alpha_0} \|D^\beta f_{\varepsilon,t}\|_{L^1(\mathbb{R}^d)} = \sum_{|\beta| \leq \alpha_0} \|D^\beta(\phi_{t\varepsilon} * f)\|_{L^1(\mathbb{R}^d)}.$$

For a fixed $|\beta| \leq \alpha_0$, since ϕ is a Schwartz function and f has compact support, we can interchange convolution and weak derivatives to obtain

$$D^\beta(\phi_{t\varepsilon} * f) = (D^{\beta-\gamma} \phi_{t\varepsilon}) * (D^\gamma f),$$

where $\gamma = \min\{|\beta|, \alpha\}$. By Young's convolution inequality ($L^1 * L^q \rightarrow L^q$) and the compact support of f , we have

$$\begin{aligned} \|D^\beta(\phi_{t\varepsilon} * f)\|_{L^1(\mathbb{R}^d)} &\leq \|D^{\beta-\gamma} \phi_{t\varepsilon}\|_{L^1(\mathbb{R}^d)} \|D^\gamma f\|_{L^1(\mathbb{R}^d)} \\ &\leq C_R \|D^{\beta-\gamma} \phi_{t\varepsilon}\|_{L^1(\mathbb{R}^d)} \|f\|_{W^{\alpha,p}(\mathbb{R}^d)}, \end{aligned}$$

where C_R comes from the compact support of f and the Sobolev embedding inequality. Direct computation gives

$$D^{\beta-\gamma} \phi_{t\varepsilon}(\mathbf{x}) = (t\varepsilon)^{-d-|\beta-\gamma|} (D^{\beta-\gamma} \phi)(\mathbf{x}/(t\varepsilon)),$$

so after the change of variables $\mathbf{x} \mapsto (t\varepsilon)\mathbf{x}$, we have

$$\begin{aligned} \|D^{\beta-\gamma} \phi_{t\varepsilon}\|_{L^1(\mathbb{R}^d)} &= (t\varepsilon)^{-d-(|\beta-\gamma|)+d} \|D^{\beta-\gamma} \phi\|_{L^1(\mathbb{R}^d)} \\ &= (t\varepsilon)^{-|\beta-\gamma|} \|D^{\beta-\gamma} \phi\|_{L^1(\mathbb{R}^d)}. \end{aligned}$$

Let $M_\phi := \max_{|\delta| \leq \alpha_0} \|D^\delta \phi\|_{L^1(\mathbb{R}^d)} < \infty$. Then

$$\begin{aligned} \|D^\beta(\phi_{t\varepsilon} * f)\|_{L^1(\mathbb{R}^d)} &\leq C_R M_\phi (t\varepsilon)^{-|\beta-\gamma|} \|f\|_{W^{\alpha,p}(\mathbb{R}^d)} \\ &= C_R M_\phi (t\varepsilon)^{-(|\beta|-\gamma)} \|f\|_{W^{\alpha,p}(\mathbb{R}^d)}. \end{aligned}$$

Therefore, summing over β , and noting that when $\alpha\varepsilon \leq 1$, $t\varepsilon \leq 1$, so $(t\varepsilon)^{-(|\beta|-\gamma)} \leq (t\varepsilon)^{-(|\beta|-\alpha)} = (t\varepsilon)^{-|\beta|+\alpha}$. We have

$$\begin{aligned} \|f_{\varepsilon,t}\|_{W^{\alpha_0,1}(\mathbb{R}^d)} &= \sum_{|\beta| \leq \alpha_0} \|D^\beta(\phi_{t\varepsilon} * f)\|_{L^1(\mathbb{R}^d)} \\ &\leq C_R M_\phi \|f\|_{W^{\alpha,p}(\mathbb{R}^d)} \sum_{|\beta| \leq \alpha_0} (t\varepsilon)^{-(|\beta|-\alpha)} \\ &= C_R M_\phi (t\varepsilon)^{-(\alpha_0-\alpha)} \|f\|_{W^{\alpha,p}(\mathbb{R}^d)} \sum_{|\beta| \leq \alpha_0} (t\varepsilon)^{|\beta|-\alpha_0}. \end{aligned}$$

Since $t\varepsilon \leq 1$, we have $(t\varepsilon)^{|\beta|-\alpha_0} \leq 1$ for all $|\beta| \leq \alpha_0$. In particular, when $t\varepsilon \leq 1$:

$$\sum_{|\beta| \leq \alpha_0} (t\varepsilon)^{|\beta|-\alpha_0} \leq \sum_{|\beta| \leq \alpha_0} 1 = \binom{\alpha_0 + d}{d}.$$

Hence,

$$\begin{aligned} \|f_{\varepsilon,t}\|_{W^{\alpha_0,1}(\mathbb{R}^d)} &\leq C_R M_\phi \binom{\alpha_0 + d}{d} (t\varepsilon)^{-(\alpha_0-\alpha)} \|f\|_{W^{\alpha,p}(\mathbb{R}^d)} \\ &\leq C_{\alpha,d} (t\varepsilon)^{-(\alpha_0-\alpha)} \|f\|_{W^{\alpha,p}(\mathbb{R}^d)}, \end{aligned}$$

where $C_{\alpha,d} = C_R M_\phi \binom{\alpha_0+d}{d} = C_R M_\phi \binom{s+2d}{d}$. From the arbitrariness of t ($t = 1, \dots, \alpha$) and the definition of f_ε in (9), we obtain

$$\begin{aligned} \|f_\varepsilon\|_{W^{\alpha_0,1}(\mathbb{R}^d)} &\leq \sum_{t=1}^{\alpha} \binom{\alpha}{t} \|f_{\varepsilon,t}\|_{W^{\alpha_0,1}(\mathbb{R}^d)} \\ &\leq C_R M_\phi \binom{s+2d}{d} \varepsilon^{-(\alpha_0-\alpha)} \|f\|_{W^{\alpha,p}(\mathbb{R}^d)} \sum_{t=1}^{\alpha} \binom{\alpha}{t} \\ &= C_R M_\phi \binom{s+2d}{d} (2^\alpha - 1) \varepsilon^{-(\alpha_0-\alpha)} \|f\|_{W^{\alpha,p}(\mathbb{R}^d)} \\ &\leq C'_{\alpha,d}(s,d) \|f\|_{W^{\alpha,p}(\mathbb{R}^d)} \varepsilon^{-(\alpha_0-\alpha)}, \end{aligned}$$

where $C'_{\alpha,d}(s,d) = C_R M_\phi \binom{s+2d}{d} (2^\alpha - 1)$ grows polynomially in d as $(2^\alpha - 1) \binom{s+2d}{d} \lesssim_s 2^{2d}$. Then, by the approximation theorem (8), there exists $f_m \in \Sigma_m^s$ such that

$$\begin{aligned} \|f_\varepsilon - f_m\|_{L^p(\mathbb{R}^d)} &\leq C_0(s,d) \|f_\varepsilon\|_{W^{\alpha_0,1}(\mathbb{R}^d)} m^{-\frac{1}{2} - \frac{s-1}{d+1}} \log m \\ &\leq C_0(s,d) \cdot C'_{\alpha,d}(s,d) \|f\|_{W^{\alpha,p}(\mathbb{R}^d)} \varepsilon^{-(\alpha_0-\alpha)} m^{-\frac{1}{2} - \frac{s-1}{d+1}} \log m, \end{aligned}$$

where $C_0(s,d)$ is the hidden constant in Theorem (8). Consequently, by the triangle inequality,

$$\begin{aligned} \|f - f_m\|_{L^p(\mathbb{R}^d)} &\leq \|f - f_\varepsilon\|_{L^p(\mathbb{R}^d)} + \|f_\varepsilon - f_m\|_{L^p(\mathbb{R}^d)} \\ &\leq C_\alpha R^\alpha \|f\|_{W^{\alpha,p}(\mathbb{R}^d)} \varepsilon^\alpha + C_0(s,d) C'_{\alpha,d}(s,d) \|f\|_{W^{\alpha,p}(\mathbb{R}^d)} \varepsilon^{-(\alpha_0-\alpha)} m^{-\frac{1}{2} - \frac{s-1}{d+1}} \log m \\ &\leq C(s,d,R,\alpha) \|f\|_{W^{\alpha,p}(\mathbb{R}^d)} \left(\varepsilon^\alpha + \varepsilon^{-(\alpha_0-\alpha)} m^{-\frac{1}{2} - \frac{s-1}{d+1}} \log m \right), \end{aligned}$$

Now choose $\varepsilon = m^{-\frac{2s+d+3}{2(s+d)(d+1)}}$ to balance the two error terms above. Substituting this yields

$$\|f - f_m\|_{L^p(\mathbb{R}^d)} \lesssim \|f\|_{W^{\alpha,p}(\mathbb{R}^d)} m^{-\frac{\alpha(2s+d+3)}{2(s+d)(d+1)}} \log m,$$

This completes the proof of Theorem 5. ■

Appendix E. Proofs in Section 4

We provide the proofs of the generalization results from Section 4.

Proof of Theorem 8

Before proving this theorem, we need some preparatory results. Define the empirical distribution associated with the sample $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ by

$$\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{\mathbf{x}_i},$$

which induces the corresponding empirical $L^2(\mathrm{d}\mu_n)$ norm

$$\|f\|_{L^2(\mathrm{d}\mu_n)}^2 = \int_{\Omega} |f(\mathbf{x})|^2 \mathrm{d}\mu_n(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n |f(\mathbf{x}_i)|^2.$$

The upper bound for the generalization error depends on the size of the chosen neural-network model class, so we need a notion of model complexity. The local complexity defined below is the main tool in the error analysis.

Definition 15 *Given a model class $\mathcal{F}$, define its local complexity by*

$$\mathcal{G}_n(\mathcal{F}, \delta, \boldsymbol{\xi}) = \mathbb{E}_{\boldsymbol{\xi}} \left[\sup_{f \in \mathcal{F}, \|f\|_{L^2(\mathrm{d}\mu_n)} \leq \delta} \left| \frac{1}{n} \sum_{i=1}^n \xi_i f(\mathbf{x}_i) \right| \right], \quad \delta > 0,$$

where $\{\xi_i\}_{i=1}^n$ is a sequence of independent identically distributed zero-mean random variables.

When $\xi_i = \eta_i$, we abbreviate it as $\mathcal{G}_n(\mathcal{F}, \delta)$; this quantity measures the correlation between the model class $B_{\partial \Sigma_m^s}(r)$ and the noise $\boldsymbol{\eta}$ on the sample $\{\mathbf{x}_i\}_{i=1}^n$. When ξ_i are Rademacher or Gaussian random variables, $\mathcal{G}_n(\mathcal{F}, \delta, \boldsymbol{\xi})$ reduces to the classical local Rademacher or local Gaussian complexity, respectively (Wainwright, 2019). The value of $\mathcal{G}_n(\delta, \boldsymbol{\xi})$ depends on the sample $\{\mathbf{x}_i\}_{i=1}^n$, but we suppress this dependence because the complexity bounds below hold uniformly over all samples.

A set is called *star-shaped* if for any f in the set and any $t \in [0, 1]$, we also have tf in the set. For a non-star-shaped set $\mathcal{F}$, we may define its star-shaped hull by

$$\text{star}(\mathcal{F}) = \{tf : f \in \mathcal{F}, t \in [0, 1]\}.$$

Lemma 16 *Let $\mathcal{F}$ be star-shaped. Viewing $\frac{\mathcal{G}_n(\mathcal{F}, \delta, \boldsymbol{\xi})}{\delta}$ as a function of δ , it is monotonically decreasing on $(0, +\infty)$. Therefore, for any constant $c > 0$,*

$$\delta_n^* = \min\{\delta > 0 : \mathcal{G}_n(\mathcal{F}, \delta, \boldsymbol{\xi}) \leq c\delta^2\}$$

always exists and is finite.

Proof Write the local complexity simply as $\mathcal{G}_n(\delta)$. For any $0 < \delta \leq t$, we show that $\mathcal{G}_n(t) \leq \frac{t}{\delta} \mathcal{G}_n(\delta)$. For any $f \in \mathcal{F}$ with $\|f\|_{L^2(\mathrm{d}\mu_n)} \leq t$, by star-shapedness we have $\frac{\delta}{t}f \in \mathcal{F}$ and $\|\frac{\delta}{t}f\|_{L^2(\mathrm{d}\mu_n)} \leq \delta$. Hence

$$\left| \frac{1}{n} \sum_{i=1}^n \xi_i f(\mathbf{x}_i) \right| = \frac{t}{\delta} \left| \frac{1}{n} \sum_{i=1}^n \xi_i \cdot \frac{\delta}{t} f(\mathbf{x}_i) \right| \leq \frac{t}{\delta} \sup_{f \in \mathcal{F}, \|f\|_{L^2(\mathrm{d}\mu_n)} \leq \delta} \left| \frac{1}{n} \sum_{i=1}^n \xi_i f(\mathbf{x}_i) \right|.$$

By the arbitrariness of f , it follows that

$$\sup_{f \in \mathcal{F}, \|f\|_{L^2(\mathrm{d}\mu_n)} \leq t} \left| \frac{1}{n} \sum_{i=1}^n \xi_i f(\mathbf{x}_i) \right| \leq \frac{t}{\delta} \sup_{f \in \mathcal{F}, \|f\|_{L^2(\mathrm{d}\mu_n)} \leq \delta} \left| \frac{1}{n} \sum_{i=1}^n \xi_i f(\mathbf{x}_i) \right|.$$

Taking expectation with respect to ξ yields the desired conclusion. This completes the proof of the lemma. $\blacksquare$

As we shall see in the subsequent proof, estimating the size of δ_n^* is the key to analyzing the upper bound of the generalization error, and for this purpose we need an upper bound on the local complexity. Dudley's entropy integral inequality (Wainwright, 2019) is a classical tool for estimating local complexity. The following theorem is in fact an application of Dudley's entropy integral inequality, and its proof relies on the classical chaining method.

Theorem 17 *Let $M > 0$, and denote by $\mathcal{B}_s(M)$ the closed ball of radius M in the s th-order Barron space. Then for any $0 < \delta \leq M$ and any independent identically distributed sub-Gaussian random variables ξ , we have*

$$\mathcal{G}_n(\mathcal{B}_s(M), \delta, \xi) \lesssim \delta^{\frac{2s+1}{d+2s+1}} M^{\frac{d}{d+2s+1}} n^{-\frac{1}{2}} \sqrt{\log \frac{nM}{\delta}}.$$

The implied constant is independent of the sample. The same upper bound also holds for the function class $\text{star}(\pi_B(\mathcal{B}_s(M)))$.

Proof For a model class $\mathcal{F}$, let $N = \mathcal{N}(\varepsilon, \mathcal{F}, \|\cdot\|_{L^2(\text{d}\mu_n)})$ denote its covering number, that is, there exist $\{f_1, \dots, f_N\} \subset \mathcal{F}$ such that for any $f \in \mathcal{F}$, there is some f_j satisfying $\|f - f_j\|_{L^2(\text{d}\mu_n)} \leq \varepsilon$. Therefore, by the Cauchy-Schwarz inequality,

$$\begin{aligned} \left| \frac{1}{n} \sum_{i=1}^n \eta_i f(\mathbf{x}_i) \right| &\leq \left| \frac{1}{n} \sum_{i=1}^n \eta_i f_j(\mathbf{x}_i) \right| + \left| \frac{1}{n} \sum_{i=1}^n \eta_i (f(\mathbf{x}_i) - f_j(\mathbf{x}_i)) \right| \\ &\leq \max_{j=1, \dots, N} \left| \frac{1}{n} \sum_{i=1}^n \eta_i f_j(\mathbf{x}_i) \right| + \sqrt{\frac{\sum_{i=1}^n \eta_i^2}{n}} \varepsilon, \end{aligned}$$

and hence, by the arbitrariness of f and taking expectation with respect to the noise,

$$\mathbb{E}_{\boldsymbol{\eta}} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \eta_i f(\mathbf{x}_i) \right| \right] \leq \mathbb{E}_{\boldsymbol{\eta}} \left[\max_{j=1, \dots, N} \left| \frac{1}{n} \sum_{i=1}^n \eta_i f_j(\mathbf{x}_i) \right| \right] + \mathbb{E} \left[\sqrt{\frac{\sum_{i=1}^n \eta_i^2}{n}} \varepsilon \right]$$

By Jensen's inequality and the moment assumption on the noise,

$$\mathbb{E} \left[\sqrt{\frac{\sum_{i=1}^n \eta_i^2}{n}} \varepsilon \right] \leq \sqrt{\mathbb{E} \left[\frac{\sum_{i=1}^n \eta_i^2}{n} \right]} \varepsilon \leq \sqrt{2CR^2} \varepsilon,$$

and therefore

$$\mathbb{E}_{\boldsymbol{\eta}} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \eta_i f(\mathbf{x}_i) \right| \right] \leq \mathbb{E}_{\boldsymbol{\eta}} \left[\max_{j=1, \dots, N} \left| \frac{1}{n} \sum_{i=1}^n \eta_i f_j(\mathbf{x}_i) \right| \right] + \sqrt{2CR^2} \varepsilon.$$

Define a family of zero-mean random variables by

$$Z(f) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \eta_i f(\mathbf{x}_i), \quad f \in \mathcal{F},$$

Then it is not hard to show that $Z(f)$ forms a sub-Gaussian process with respect to the metric $\rho_Z(f, f') = \|f - f'\|_{L^2(\text{d}\mu_n)}$, namely, for any $f, f' \in \mathcal{F}$,

$$\|Z(f) - Z(f')\|_{\psi_2} \leq K \|f - f'\|_{L^2(\text{d}\mu_n)},$$

Hence, by the chaining method (Wainwright, 2019, Theorem 5.22),

$$\mathbb{E}_{\boldsymbol{\eta}} \left[\max_{j=1, \dots, N} \left| \frac{1}{n} \sum_{i=1}^n \eta_i f_j(\mathbf{x}_i) \right| \right] \leq \frac{16}{\sqrt{n}} \int_{\frac{\varepsilon}{4}}^{\frac{D}{2}} \sqrt{\log \mathcal{N}(u, \mathcal{F}, \|\cdot\|_{L^2(\mathrm{d}\mu_n)})} du.$$

Here $D = \sup_{f, f' \in \mathcal{F}} \rho_Z(f, f')$ denotes the diameter of $\mathcal{F}$. Combining the above and using the arbitrariness of ε , we obtain

$$\mathbb{E}_{\boldsymbol{\eta}} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \eta_i f(\mathbf{x}_i) \right| \right] \leq \inf_{\varepsilon \geq 0} \left\{ 4\sqrt{2CR^2}\varepsilon + \frac{16}{\sqrt{n}} \int_{\varepsilon}^{\frac{D}{2}} \sqrt{\log \mathcal{N}(u, \mathcal{F}, \|\cdot\|_{L^2(\mathrm{d}\mu_n)})} du \right\}.$$

Now set $\mathcal{F} = \{f \in \mathcal{B}_s(M) : \|f\|_{L^2(\mathrm{d}\mu_n)} \leq \delta\}$, so that $D = 2\delta$. It remains only to estimate its covering number. By following the proof of (Yang and Zhou, 2024, Theorem 5), one obtains

$$\log \mathcal{N}(\varepsilon, \mathcal{B}_s(M), \|\cdot\|_{L^2(\mathrm{d}\mu_n)}) \lesssim \left(\frac{\varepsilon}{M} \right)^{-\frac{2d}{d+2s+1}} \log \left(1 + \frac{M}{\varepsilon} \right).$$

Therefore,

$$\mathcal{G}_n(\mathcal{B}_s(M), \delta) \lesssim \inf_{\varepsilon \geq 0} \left\{ 4\sqrt{2CR^2}\varepsilon + \frac{16}{\sqrt{n}} \int_{\varepsilon}^{\delta} \left(\frac{u}{M} \right)^{-\frac{d}{d+2s+1}} \sqrt{\log \left(1 + \frac{M}{u} \right)} du \right\}$$

Taking $\varepsilon \asymp \delta^{\frac{2s+1}{d+2s+1}} M^{\frac{d}{d+2s+1}} n^{-\frac{1}{2}}$, we get

$$\begin{aligned} \mathcal{G}_n(\mathcal{B}_s(M), \delta) &\lesssim \delta^{\frac{2s+1}{d+2s+1}} M^{\frac{d}{d+2s+1}} n^{-\frac{1}{2}} + M^{\frac{d}{d+2s+1}} n^{-\frac{1}{2}} \sqrt{\log \left(1 + \frac{M}{\varepsilon} \right)} \int_0^{\delta} u^{-\frac{d}{d+2s+1}} du \\ &\lesssim \delta^{\frac{2s+1}{d+2s+1}} M^{\frac{d}{d+2s+1}} n^{-\frac{1}{2}} \sqrt{\log \frac{nM}{\delta}}. \end{aligned}$$

Finally, we prove the case $\mathcal{F} = \text{star}(\pi_B(\mathcal{B}_s(M)))$. By the Lipschitz continuity of π_B , we have $\mathcal{N}(\delta, \pi_B(\mathcal{B}_s(M))) \lesssim \mathcal{N}(\delta, \mathcal{B}_s(M))$. On the other hand, let $\{f_1, \dots, f_m\}$ be a $\frac{\delta}{2}$ -cover of $\pi_B(\mathcal{B}_s(M))$, and let $\{t_1, \dots, t_L\}$ be a $\frac{\delta}{2B}$ -cover of $[0, 1]$. Then for any $f \in \text{star}(\pi_B(\mathcal{B}_s(M)))$, there exist $t \in [0, 1]$ and $g \in \pi_B(\mathcal{B}_s(M))$ such that $f = tg$. By the definition of the covering sets, there exist t_j, g_i such that $|t - t_j| \leq \frac{\delta}{2B}$ and $\|g - g_i\|_{L^2(\mathrm{d}\mu_n)} \leq \frac{\delta}{2}$. Hence

$$\begin{aligned} \|f - t_j g_i\|_{L^2(\mathrm{d}\mu_n)} &= \|tg - t_j g_i\|_{L^2(\mathrm{d}\mu_n)} \\ &\leq \|t(g - g_i)\|_{L^2(\mathrm{d}\mu_n)} + \|(t - t_j)g_i\|_{L^2(\mathrm{d}\mu_n)} \\ &\leq \|g - g_i\|_{L^2(\mathrm{d}\mu_n)} + |t - t_j| \cdot \|g_i\|_{L^2(\mathrm{d}\mu_n)} \\ &\leq \frac{\delta}{2} + B \cdot \frac{\delta}{2B} = \delta. \end{aligned}$$

This shows that

$$\begin{aligned} \log \mathcal{N}(\delta, \text{star}(\pi_B(\mathcal{B}_s(M))), \|\cdot\|_{L^2(\mathrm{d}\mu_n)}) &\lesssim \left(\frac{\delta}{M} \right)^{-\frac{d}{d+2s+1}} \log \left(1 + \frac{M}{\delta} \right) + \log \left(1 + \frac{B}{\delta} \right) \\ &\lesssim \left(\frac{\delta}{M} \right)^{-\frac{d}{d+2s+1}} \log \left(1 + \frac{M}{\delta} \right). \end{aligned}$$

The conclusion follows by the same argument as above. This completes the proof of Theorem 17. $\blacksquare$

Finally, we introduce a quantity finer than complexity, which arises from the generic chaining method (Talagrand, 2021). Consider a set T equipped with a seminorm d . Let $\{A_k\}_{k=0}^\infty$ be an increasing sequence of partitions of T , meaning that each A_i is a partition of T , and every element in A_{k+1} is contained in some element of A_k . If $|A_0| = 1$ and for every $k \geq 1$ we have $|A_k| \leq 2^{2^k}$, then $\{A_k\}_{k=0}^\infty$ is called an *admissible sequence* of T . For $t \in T$, we write $A_n(t)$ for the unique element of A_n containing t . Define

$$\gamma_2(T, d) = \inf \sup_{t \in T} \sum_{k=0}^{\infty} 2^{\frac{k}{2}} \text{diam}(A_k(t)),$$

where the infimum is taken over all admissible sequences of T . This quantity is closely related to Dudley's entropy integral; in fact,

$$\gamma_2(T, d) \lesssim \int_0^{\text{diam}(T)} \sqrt{\log \mathcal{N}(u, T, d)} du.$$

Moreover, if $T = \{f \in \mathcal{F} : \|f\|_{L^2(\text{d}\mu_n)} \leq \delta\}$ and d is induced by $\|\cdot\|_{L^2(\text{d}\mu_n)}$, then when $\boldsymbol{\eta}$ is Gaussian, Talagrand's theorem states that $\mathcal{G}_n(\mathcal{F}, \delta, \boldsymbol{\eta}) \asymp \gamma_2(T, d)$.

Consider $T(\delta) = \{f \in \text{star}(\partial(\Sigma_m^s)) : \|f\|_{L^2(\text{d}\mu_n)} \leq \delta\}$ with d induced by $\|\cdot\|_{L^2(\text{d}\mu_n)}$. Denote

$$\gamma_2(\delta) = \gamma_2(T(\delta), d).$$

Since $\text{star}(\partial(\Sigma_m^s))$ is star-shaped and by the properties of γ_2 , it is easy to show that $\frac{\gamma_2(\delta)}{\delta}$ is monotonically decreasing on $(0, +\infty)$.

Before formally deriving the upper bound on the error, we first briefly describe the proof strategy. We begin with a decomposition of the excess error. Let $B = B(m) > \sup_{x \in \Omega} |f_\rho(x)| > 0$ be a truncation constant to be determined. For any $f \in \Sigma_m^s$, consider

$$\begin{aligned} \|\pi_B f_m - f_\rho\|_{L^2(\text{d}\mu)}^2 &\leq 2\|\pi_B f_m - \pi_B f\|_{L^2(\text{d}\mu)}^2 + 2\|\pi_B f - f_\rho\|_{L^2(\text{d}\mu)}^2 \\ &\leq 2\|\pi_{2B}(f_m - f)\|_{L^2(\text{d}\mu)}^2 + 2\|f - f_\rho\|_{L^2(\text{d}\mu)}^2, \end{aligned}$$

where we have used $|\pi_B f_m - \pi_B f| \leq |\pi_{2B}(f_m - f)|$ and $B > \sup_{x \in \Omega} |f_\rho(x)|$. The second term on the right-hand side can be controlled directly by the approximation theorem 5. For the first term, following the method of (Wainwright, 2019, Chapter 13, Chapter 14), we first estimate the empirical error $\|\pi_{2B}(f_m - f)\|_{L^2(\text{d}\mu_n)}^2$ under a fixed-design setting, and then use a uniform law of large numbers to pass from the fixed-design setting to the random-design setting, thereby obtaining an upper bound on $\|\pi_{2B}(f_m - f)\|_{L^2(\text{d}\mu)}^2$. For the empirical error $\|\pi_{2B}(f_m - f)\|_{L^2(\text{d}\mu_n)}^2$, note that for any $f \in \Sigma_m^s$,

$$\|\pi_{2B}(f_m - f)\|_{L^2(\text{d}\mu_n)}^2 \leq \|f_m - f\|_{L^2(\text{d}\mu_n)}^2 \leq 2\|f_m - f_\rho\|_{L^2(\text{d}\mu_n)}^2 + 2\|f - f_\rho\|_{L^2(\text{d}\mu_n)}^2,$$

Again, the second term on the right-hand side can be controlled directly by Theorem 5, so it remains only to estimate the empirical error $\|f_m - f_\rho\|_{L^2(\text{d}\mu_n)}^2$ of the empirical risk minimizer f_m , which is precisely the core of the upper-bound analysis. We now carry out the complete proof in three steps.

STEP 1: UPPER BOUNDS ON THE EMPIRICAL ERROR AND THE ℓ^1 PATH NORM

In this subsection, we fix the sample $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ and analyze upper bounds for the empirical error and the ℓ^1 path norm of the empirical risk minimizer f_m . For any $f \in \Sigma_m^s$, by definition,

$$\frac{1}{n} \sum_{i=1}^n (f_m(\mathbf{x}_i) - y_i)^2 + \lambda \|f_m\|_{\mathcal{P}_{1,s}} \leq \frac{1}{n} \sum_{i=1}^n (f(\mathbf{x}_i) - y_i)^2 + \lambda \|f\|_{\mathcal{P}_{1,s}}.$$

Noting that $y_i = f_\rho(\mathbf{x}_i) + \eta_i$, substituting this and rearranging yields the *basic inequality*

$$\|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f_m\|_{\mathcal{P}_{1,s}} \leq \|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} + \frac{2}{n} \sum_{i=1}^n \eta_i (f_m(\mathbf{x}_i) - f(\mathbf{x}_i)).$$

From this basic inequality and the oracle inequality to be derived below, we can obtain upper bounds for both the empirical error and the ℓ^1 path norm. To state and prove this oracle inequality, we need a tail bound for the Gaussian empirical process.

Theorem 18 *Let $M, \delta > 0$ and define the sub-Gaussian empirical process*

$$Z(\delta) = \sup_{\substack{f \in \text{star}(\partial(\Sigma_m^s)), \\ \|f\|_{L^2(\mathrm{d}\mu_n)} \leq \delta}} \left| \frac{1}{n} \sum_{i=1}^n \eta_i f(\mathbf{x}_i) \right|,$$

Then there exist constants $L, L' > 0$ depending only on the Gaussian constant σ such that Z satisfies the concentration inequality

$$\mathbb{P} \left[Z(\delta) \geq \frac{L}{\sqrt{n}} \gamma(\delta) + t \right] \leq L' e^{-\frac{nt^2}{4L^2\delta^2}} \quad \forall t > 0.$$

Yang and Zhou (2024) established generalization results under the ℓ_2 path norm only for bounded noise. For bounded noise or Gaussian noise, one may directly use the classical Hoeffding inequality or Gaussian concentration inequality to obtain similar conclusions. Here we provide the proof under sub-Gaussian noise. In this case, the previous two concentration inequalities are no longer applicable, so we introduce the finer quantity $\gamma(\delta)$ to characterize the concentration property of the empirical process.

Lemma 19 *Let $c_0 \geq 1$ be a constant, and suppose f_m is the empirical risk minimizer such that for any $f \in \Sigma_m^s$,*

$$\|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f_m\|_{\mathcal{P}_{1,s}} \leq c_0 \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} \right) + \frac{2}{n} \left| \sum_{i=1}^n \eta_i (f_m(\mathbf{x}_i) - f(\mathbf{x}_i)) \right|.$$

For a user-defined parameter $M > 0$, let $\delta_n = \delta_n(M)$ satisfy

$$\frac{L}{\sqrt{n}} \gamma(\delta_n) \leq \delta_n^2.$$

Then there exist constants $c_1, L' > 0$ such that if $\lambda \geq \frac{8\delta_n^2}{M}$, then

$$\|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f_m\|_{\mathcal{P}_{1,s}} \leq (1 + 2c_0) \inf_{f \in \Sigma_m^s} \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} \right) + 64\delta_n^2 \quad (10)$$

with probability at least $1 - L' e^{-\frac{n\delta_n^2}{4L^2}}$, where the implied constants are independent of the sample.

Proof For any $f \in \Sigma_m^s$, write $g_f = f_m - f$. Then $\|g_f\|_{\mathcal{P}_{1,s}} \leq \|f_m\|_{\mathcal{P}_{1,s}} + \|f\|_{\mathcal{P}_{1,s}}$. We divide the discussion into four cases.

Case 1: $\|g\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|g_f\|_{\mathcal{P}_{1,s}} \leq \delta_n^2$. Then

$$\begin{aligned} \|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f_m\|_{\mathcal{P}_{1,s}} &\leq 2\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + 2\|g_f\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} + \lambda \|g_f\|_{\mathcal{P}_{1,s}} \\ &\leq 2\|f - h\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} + 2\delta_n^2. \end{aligned}$$

Case 2: $\|g_f\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda\|g_f\|_{\mathcal{P}_{1,s}} > \delta_n^2$ and $\|f_m\|_{\mathcal{P}_{1,s}} \leq M$. In this case $g_f \in \partial\Sigma_m^s$. Consider the event

$$A_1 = \left\{ \exists g \in \partial\Sigma_m^s : \|g\|_{L^2(\mathrm{d}\mu_n)} + \lambda\|g\|_{\mathcal{P}_{1,s}} > \delta_n^2, \left| \frac{1}{n} \sum_{i=1}^n \eta_i g(\mathbf{x}_i) \right| > 2\delta_n \sqrt{\|g\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda\|g\|_{\mathcal{P}_{1,s}}} \right\}.$$

If A_1 occurs, then there exists

$$\tilde{g} = \frac{\delta_n}{\sqrt{\|g\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda\|g\|_{\mathcal{P}_{1,s}}}} g \in \text{star}(\partial\Sigma_m^s)$$

such that $\|\tilde{g}\|_{L^2(\mathrm{d}\mu_n)} \leq \delta_n$ and

$$\left| \frac{1}{n} \sum_{i=1}^n \eta_i \tilde{g}(\mathbf{x}_i) \right| = \frac{\delta_n}{\sqrt{\|g\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda\|g\|_{\mathcal{P}_{1,s}}}} \left| \frac{1}{n} \sum_{i=1}^n \eta_i g(\mathbf{x}_i) \right| > 2\delta_n^2.$$

This implies that $\mathbb{P}[A_1] \leq \mathbb{P}[Z_n(\delta_n) > 2\delta_n^2]$, where

$$Z_n(\delta_n) = \sup_{\substack{g \in \text{star}(\partial\Sigma_m^s), \\ \|g\|_{L^2(\mathrm{d}\mu_n)} \leq \delta_n}} \left| \frac{1}{n} \sum_{i=1}^n \eta_i g(\mathbf{x}_i) \right|.$$

is a sub-Gaussian empirical process. By Theorem 18,

$$\mathbb{P} \left[Z_n(\delta_n) \geq \frac{L}{\sqrt{n}} \gamma(\delta_n) + t \right] \leq L' e^{-\frac{nt^2}{4L^2\delta_n^2}}.$$

Since by assumption $\frac{L}{\sqrt{n}} \gamma(\delta_n) \leq \delta_n^2$, taking $t = \delta_n^2$ yields

$$\mathbb{P}[A_1] \leq \mathbb{P}[Z_n(\delta_n) > 2\delta_n^2] \leq \mathbb{P}[Z_n(\delta_n) \geq L\gamma(\delta_n) + \delta_n^2] \leq L' e^{-\frac{n\delta_n^2}{4L^2}}.$$

This shows that $\overline{A}_1$ holds with high probability; specifically,

$$\mathbb{P} \left[\left| \frac{1}{n} \sum_{i=1}^n \eta_i g_f(\mathbf{x}_i) \right| \leq 2\delta_n \sqrt{\|g_f\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda\|g_f\|_{\mathcal{P}_{1,s}}} \right] \geq 1 - L' e^{-\frac{n\delta_n^2}{4L^2}}.$$

Combining this with the assumed inequality, it follows that with at least the same probability,

$$\begin{aligned} & \|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda\|f_m\|_{\mathcal{P}_{1,s}} \\ & \leq c_0 \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda\|f\|_{\mathcal{P}_{1,s}} \right) + 4\delta_n \sqrt{\|g_f\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda\|g_f\|_{\mathcal{P}_{1,s}}} \\ & \leq c_0 \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda\|f\|_{\mathcal{P}_{1,s}} \right) + 16\delta_n^2 + \frac{1}{4}\|f_m - f\|_{L^2(\mathrm{d}\mu_n)}^2 + \frac{1}{4}\lambda\|f_m - f\|_{\mathcal{P}_{1,s}} \\ & \leq c_0 \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda\|f\|_{\mathcal{P}_{1,s}} \right) + 16\delta_n^2 + \frac{1}{2}\|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \frac{1}{2}\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 \\ & \quad + \frac{1}{4}\lambda\|f_m\|_{\mathcal{P}_{1,s}} + \frac{1}{4}\lambda\|f\|_{\mathcal{P}_{1,s}} \\ & \leq \left(\frac{1}{2} + c_0 \right) \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda\|f\|_{\mathcal{P}_{1,s}} \right) + 16\delta_n^2 + \frac{1}{2} \left(\|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda\|f_m\|_{\mathcal{P}_{1,s}} \right), \end{aligned}$$

or equivalently,

$$\|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f_m\|_{\mathcal{P}_{1,s}} \leq (1 + 2c_0) \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} \right) + 32\delta_n^2.$$

Case 3: $\|f_m\|_{\mathcal{P}_{1,s}} > M$ and $\|g_f\|_{L^2(\mathrm{d}\mu_n)} \leq \frac{\delta_n \|f_m\|_{\mathcal{P}_{1,s}}}{M}$. In this case,

$$\begin{aligned} \tilde{g}_f &= \frac{R}{\|f_m\|_{\mathcal{P}_{1,s}}} g_f = \frac{M}{\|f_m\|_{\mathcal{P}_{1,s}}} (f_m - f) \in \partial\Sigma_m^s, \\ \|\tilde{g}_f\|_{L^2(\mathrm{d}\mu_n)} &= \frac{M}{\|f_m\|_{\mathcal{P}_{1,s}}} \|g_f\|_{L^2(\mathrm{d}\mu_n)} \leq \delta_n. \end{aligned}$$

By Case 2, $\mathbb{P}[Z_n(\delta_n) > 2\delta_n^2] \leq L'e^{-\frac{n\delta_n^2}{4L^2}}$. Therefore,

$$\left| \frac{1}{n} \sum_{i=1}^n \eta_i g_f(\mathbf{x}_i) \right| = \frac{\|f_m\|_{\mathcal{P}_{1,s}}}{M} \left| \frac{1}{n} \sum_{i=1}^n \eta_i \tilde{g}_f(\mathbf{x}_i) \right| \leq \frac{2\delta_n^2 \|f_m\|_{\mathcal{P}_{1,s}}}{M}$$

holds with probability at least $1 - L'e^{-\frac{n\delta_n^2}{4L^2}}$. Combining this with the basic inequality, we find that with at least the same probability,

$$\begin{aligned} &\|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f_m\|_{\mathcal{P}_{1,s}} \\ &\leq c_0 \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} \right) + \frac{4\delta_n^2}{M} \|f_m\|_{\mathcal{P}_{1,s}} \\ &\leq c_0 \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} \right) + \frac{\lambda}{2} \|f_m\|_{\mathcal{P}_{1,s}}, \end{aligned}$$

where the last step uses $\lambda \geq \frac{8\delta_n^2}{M}$. Hence,

$$\|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f_m\|_{\mathcal{P}_{1,s}} \leq 2c_0 \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} \right).$$

Case 4: $\|f_m\|_{\mathcal{P}_{1,s}} > M$ and $\|g_f\|_{L^2(\mathrm{d}\mu_n)} > \frac{\delta_n \|f_m\|_{\mathcal{P}_{1,s}}}{M}$. Under this assumption,

$$\tilde{g}_f = \frac{M}{\|f_m\|_{\mathcal{P}_{1,s}}} g_f \in \partial\Sigma_m^s, \quad \|\tilde{g}_f\|_{L^2(\mathrm{d}\mu_n)} > \delta_n.$$

We show that the event

$$A_2 = \left\{ \exists g \in \partial\Sigma_m^s : \|g\|_{L^2(\mathrm{d}\mu_n)} > \delta_n, \left| \frac{1}{n} \sum_{i=1}^n \eta_i g(\mathbf{x}_i) \right| > 2\delta_n \|g\|_{L^2(\mathrm{d}\mu_n)} + \frac{1}{16} \|g\|_{L^2(\mathrm{d}\mu_n)}^2 \right\}$$

occurs with small probability. To this end, for $k = 0, 1, \dots$ define $t_k = 2^k \delta_n$ and the event

$$A_2^k = \left\{ \exists g \in \partial\Sigma_m^s : t_k < \|g\|_{L^2(\mathrm{d}\mu_n)} \leq 2t_k, \left| \frac{1}{n} \sum_{i=1}^n \eta_i g(\mathbf{x}_i) \right| > 2\delta_n \|g\|_{L^2(\mathrm{d}\mu_n)} + \frac{1}{16} \|g\|_{L^2(\mathrm{d}\mu_n)}^2 \right\}.$$

Then $A_2 = \bigcup_{k=0}^{\infty} A_2^k$. If A_2^k occurs, then there exists $g \in \partial\Sigma_m^s$ such that $t_k < \|g\|_{L^2(\mathrm{d}\mu_n)} \leq t_{k+1}$ and

$$\begin{aligned} \left| \frac{1}{n} \sum_{i=1}^n \eta_i g(\mathbf{x}_i) \right| &> 2\delta_n \|g\|_{L^2(\mathrm{d}\mu_n)} + \frac{1}{16} \|g\|_{L^2(\mathrm{d}\mu_n)}^2 \\ &\geq 2\delta_n t_k + \frac{1}{16} t_k^2 = t_{k+1} \delta_n + \frac{1}{64} t_{k+1}^2, \end{aligned}$$

and therefore

$$\mathbb{P}[A_2^k] \leq \mathbb{P}\left[Z_n(t_{k+1}) > t_{k+1}\delta_n + \frac{1}{64}t_{k+1}^2\right].$$

By Theorem 18,

$$\mathbb{P}\left[Z_n(t_{k+1}) \geq \frac{L}{\sqrt{n}}\gamma(t_{k+1}) + t\right] \leq L'e^{-\frac{nt^2}{4L^2t_{k+1}^2}}.$$

Since $\text{star}(\partial\Sigma_m^s)$ is star-shaped, $\frac{\gamma(t)}{t}$ is decreasing in r , and thus for any $t \geq \delta_n$,

$$\frac{L\gamma(t)}{\sqrt{n}} \leq t \frac{L\gamma(\delta_n)}{\delta_n\sqrt{n}} \leq t\delta_n,$$

hence setting $t = \frac{1}{64}t^2$ yields

$$\mathbb{P}\left[Z_n(t) > t\delta_n + \frac{1}{64}t^2\right] \leq L'e^{-\frac{nt^2}{4^7L^2}}.$$

By countable additivity of probability,

$$\mathbb{P}[A_2] \leq \sum_{k=0}^{\infty} \mathbb{P}[A_2^k] \leq \sum_{k=0}^{\infty} L'e^{-\frac{nt_k^2}{4^7L^2}} = L' \sum_{k=0}^{\infty} e^{-\frac{n\delta_n^2}{4^7L^2}4^k} \leq c_1 e^{-\frac{n\delta_n^2}{4^7L^2}}.$$

This shows that

$$\left|\frac{1}{n} \sum_{i=1}^n \eta_i g_f(\mathbf{x}_i)\right| \leq 2\delta_n \|g_f\|_{L^2(\mathrm{d}\mu_n)} + \frac{1}{16} \|g_f\|_{L^2(\mathrm{d}\mu_n)}^2$$

holds with probability at least $1 - c_1 e^{-\frac{n\delta_n^2}{4^7L^2}}$. Multiplying both sides by $\frac{\|f_m\|_{L^2(\mathrm{d}\mu_n)}}{M}$, we get

$$\begin{aligned} \left|\frac{1}{n} \sum_{i=1}^n \eta_i g_f(\mathbf{x}_i)\right| &\leq 2\delta_n \|g_f\|_{L^2(\mathrm{d}\mu_n)} + \frac{1}{16} \frac{M}{\|f_m\|_{L^2(\mathrm{d}\mu_n)}} \|g_f\|_{L^2(\mathrm{d}\mu_n)}^2 \\ &\leq 2\delta_n \|g_f\|_{L^2(\mathrm{d}\mu_n)} + \frac{1}{16} \|g_f\|_{L^2(\mathrm{d}\mu_n)}^2, \end{aligned}$$

where the last step uses the assumption $\|f_m\|_{\mathcal{P}_{1,s}} > M$. Combining this with the basic inequality, we find that with at least the same probability,

$$\begin{aligned} &\|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f_m\|_{\mathcal{P}_{1,s}} \\ &\leq c_0 \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} \right) + 4\delta_n \|g_f\|_{L^2(\mathrm{d}\mu_n)} + \frac{1}{8} \|g_f\|_{L^2(\mathrm{d}\mu_n)}^2 \\ &\leq c_0 \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} \right) + 32\delta_n^2 + \frac{1}{4} \|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \frac{1}{4} \|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 \\ &\leq c_0 \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} \right) + 32\delta_n^2 + \frac{1}{2} \|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \frac{1}{2} \|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 \\ &\leq \left(\frac{1}{2} + c_0 \right) \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} \right) + 32\delta_n^2 + \frac{1}{2} \left(\|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f_m\|_{\mathcal{P}_{1,s}} \right), \end{aligned}$$

and rearranging gives

$$\|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f_m\|_{\mathcal{P}_{1,s}} \leq (1 + 2c_0) \left(\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}} \right) + 64\delta_n^2.$$

This completes the proof of Lemma 19. ■

Lemma 19 is in fact an oracle inequality. Equation (10) shows that the empirical error of the empirical risk minimizer f_m can be decomposed into two parts: one is the approximation error $\inf_{f \in \Sigma_m^s} (\|f - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}})$, and the other is the complexity term $64\delta_n^2$ of the model class. These two terms can be controlled respectively by the approximation theorem 5 and tools from statistical learning theory.

Equation (10) also provides an upper bound for the ℓ^1 path norm. As an example, suppose that Y has a uniform bound $B > 0$ (in which case $\boldsymbol{\eta}$ is bounded and hence sub-Gaussian). In general, taking $f = 0$ in the basic inequality yields

$$\|f_m\|_{\mathcal{P}_{1,s}} \leq \frac{1}{m\lambda} \sum_{i=1}^m y_i^2 \leq \frac{B^2}{\lambda},$$

but this estimate is quite crude. In fact, if we define

$$f^* = \arg \min_{f \in \Sigma_n^s} \|f - f_\rho\|_{L^2(\mathrm{d}\mu)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}},$$

then by definition,

$$\|f^*\|_{\mathcal{P}_{1,s}} \leq \frac{\inf_{f \in \Sigma_n^s} (\|f - f_\rho\|_{L^2(\mathrm{d}\mu)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}})}{\lambda}.$$

Since f_m is an approximation of f^* , we expect $\|f_m\|_{\mathcal{P}_{1,s}}$ to admit a similar upper bound, and this upper bound is much smaller than $\frac{B^2}{\lambda}$. Indeed, if λ is chosen appropriately (for example, so that $\lim_{m \rightarrow \infty} \lambda(m) = 0$), then $\inf_{f \in \Sigma_n^s} (\|f - f_\rho\|_{L^2(\mathrm{d}\mu)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}}) \rightarrow 0$ while B^2 is a fixed constant. This phenomenon has been studied in (Steinwart and Christmann, 2008; Shi et al., 2011, 2019), where iterative methods were proposed to improve this upper bound. However, using the oracle inequality above, we can directly obtain a better upper bound. Specifically, if we take $\lambda = \frac{8\delta_n^2}{M}$, then with high probability

$$\|f_m\|_{\mathcal{P}_{1,s}} \lesssim \frac{\inf_{f \in \Sigma_m^s} \|f - f_\rho\|_{L^\infty(\Omega)}^2}{\lambda} + M,$$

which is already very close to $\frac{\inf_{f \in \Sigma_n^s} (\|f - f_\rho\|_{L^2(\mathrm{d}\mu)}^2 + \lambda \|f\|_{\mathcal{P}_{1,s}})}{\lambda}$.

STEP 2: LOCALIZATION AND THE UNIFORM LAW OF LARGE NUMBERS

In the previous subsection, we treated the sample $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ as fixed and obtained an upper bound on the empirical error. However, what we are actually concerned with is the generalization error. In this subsection, we regard the sample as random and use the localization technique and uniform law of large numbers introduced in (Wainwright, 2019, Chapter 14) to analyze the upper bound on the generalization error.

Since the $\mathbf{x}_i$ are independent and identically distributed, for any square-integrable f we have

$$\mathbb{E}[\|f\|_{L^2(\mathrm{d}\mu_n)}^2] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n f(\mathbf{x}_i)^2\right] = \mathbb{E}[f(x)^2] = \|f\|_{L^2(\mathrm{d}\mu)}^2.$$

Therefore, by the law of large numbers, under suitable conditions $\|f\|_{L^2(\mathrm{d}\mu_n)}$ converges to $\|f\|_{L^2(\mathrm{d}\mu)}$. For example, if f is bounded, then Hoeffding's inequality (Wainwright, 2019, Proposition 2.5) yields

$$\mathbb{P}\left[\left|\|f\|_{L^2(\mathrm{d}\mu_n)}^2 - \|f\|_{L^2(\mathrm{d}\mu)}^2\right| \geq t\right] \leq 2e^{-cnt^2}, \quad \forall t > 0.$$

Using the uniform law of large numbers, one can make this tail bound hold uniformly over function classes satisfying certain conditions. The following theorem is taken from (Wainwright, 2019, Theorem 14.1, Proposition 14.25) or (Yang and Zhou, 2024, Lemma 7).

Theorem 20 *Let the function class $\mathcal{F}$ be star-shaped and uniformly bounded by some constant $B > 0$. Let ε_n satisfy*

$$\mathcal{G}_n(\mathcal{F}, \varepsilon_n, \varepsilon) \leq \frac{\varepsilon_n^2}{B},$$

where ε denotes independent identically distributed Rademacher random variables. Then there exist constants $c_1, c_2 > 0$ such that

$$\|f\|_{L^2(\mathrm{d}\mu)}^2 \leq 2\|f\|_{L^2(\mathrm{d}\mu_n)}^2 + \varepsilon_n^2, \quad \forall f \in \mathcal{F}$$

with probability at least $1 - c_1 e^{-\frac{c_2 n \varepsilon_n^2}{B^2}}$.

Since for any $f \in \Sigma_m, M^s$ we have

$$\pi_{2B}(f_m - f) \in \text{star}(\pi_{2B}(\Sigma_{2n, \|f_m\|_{\mathcal{P}_{1,s}}^s + M)) \subset \text{star}(\pi_{2B}(\mathcal{B}_s(\|f_m\|_{\mathcal{P}_{1,s}} + M))),$$

the theorem above implies that

$$\|\pi_{2B}(f_m - f)\|_{L^2(\mathrm{d}\mu)}^2 \leq 2\|\pi_{2B}(f_m - f)\|_{L^2(\mathrm{d}\mu_n)}^2 + \varepsilon_n^2, \quad \forall f \in \Sigma_{m,M}^s$$

holds with high probability.

STEP 3: COMBINING THE ABOVE ANALYSIS TO COMPLETE THE UPPER BOUND FOR THE GENERALIZATION ERROR

Proof of Theorem 8. We first determine $\delta_n = \delta_n(M)$ satisfying $\frac{L\gamma(\delta_n)}{\sqrt{n}} \leq \delta_n^2$, where $M > 0$ is to be chosen. Without loss of generality, assume that n is sufficiently large so that $\delta_n \leq M$. Since $\text{star}(\partial\Sigma_m^s) \subset \mathcal{B}_s(2M)$, we have

$$\begin{aligned} \frac{L\gamma(\delta_n)}{\sqrt{n}} &\lesssim \frac{1}{\sqrt{n}} \int_0^{2\delta_n} \sqrt{\log \mathcal{N}(u, \mathcal{B}_s(2M), \|\cdot\|_{L^2(\mathrm{d}\mu_n)})} du \\ &\leq \frac{1}{\sqrt{n}} \int_0^{2\delta_n} \sqrt{\log \mathcal{N}(u, \mathcal{B}_s(2M), \|\cdot\|_{L^2(\Omega)})} du \\ &\lesssim \frac{1}{\sqrt{n}} \int_0^{\delta_n} \left(\frac{u}{M}\right)^{-\frac{d}{d+2s+1}} \sqrt{\log\left(1 + \frac{M}{u}\right)} du \\ &\lesssim \frac{M^{\frac{d}{d+2s+1}}}{\sqrt{n}} \int_0^{\delta_n} u^{-\frac{d}{d+2s+1}} \left(\sqrt{\log 2M} + \sqrt{\log \frac{1}{u}}\right) du \\ &\lesssim \delta_n^{\frac{2s+1}{d+2s+1}} M^{\frac{d}{d+2s+1}} n^{-\frac{1}{2}} \sqrt{\log \frac{nM}{\delta_n}}. \end{aligned}$$

Thus we may take

$$\delta_n^2 \asymp n^{-\frac{d+2s+1}{2d+2s+1}} M^{\frac{2d}{2d+2s+1}} \log nM,$$

and denote the approximation error of Σ_m^s by

$$\mathcal{E}(M) = \inf_{f \in \Sigma_m^s} \|f - f_\rho\|_{L^\infty(\Omega)}^2,$$

By Lemma 19, taking $\lambda = \frac{8\delta_n^2}{M}$ yields

$$\|f_m - f_\rho\|_{L^2(\mathrm{d}\mu_n)}^2 + \lambda \|f_m\|_{\mathcal{P}_{1,s}} \lesssim \mathcal{E}(M) + \delta_n^2,$$

and in particular,

$$\|f_m\|_{\mathcal{P}_{1,s}} \lesssim \frac{\mathcal{E}(M) + \delta_n^2}{\lambda} \lesssim \frac{\mathcal{E}(M)}{\lambda} + M.$$

By the approximation theorem 5, there exists $f_m \in \Sigma_m^s$ such that

$$\|f_m - f_\rho\|_{L^\infty(\Omega)} \lesssim \begin{cases} m^{-\frac{1}{2} - \frac{2s+1}{2d}}, & \alpha \geq s + \frac{d+1}{2} \\ m^{-\frac{\alpha}{d}}, & \alpha \in \mathbb{N} \cap (0, s + \frac{d+1}{2}) \end{cases},$$

but in order to estimate $\mathcal{E}(M)$, we also need to estimate the ℓ^1 path norm $\|f_m\|_{\mathcal{P}_{1,s}}$ of the network f_m . From the proof of (Mao et al., 2026; Siegel and Xu, 2024), we know that we can find $f_m(\mathbf{x}) = \frac{1}{m} \sum_{i=1}^m m a_i \sigma_s(\mathbf{w}_i \cdot \mathbf{x} + b_i)$ satisfying $\sum_{i=1}^m |a_i| \leq \widetilde{M}$, $\mathbf{w}_i \in \mathbb{S}^{d-1}$, $b_i \in [-\sqrt{d}, \sqrt{d}]$, where

$$\widetilde{M} = \begin{cases} C m^{\frac{2s+d+1-2\alpha}{2d}}, & \alpha < s + \frac{d+1}{2} \text{ and } \alpha \in \mathbb{N} \\ 1, & \alpha \geq s + \frac{d+1}{2} \end{cases},$$

and therefore it is easy to estimate that the ℓ^1 path norm of such an f_m satisfies

$$\|f_m\|_{\mathcal{P}_{1,s}} \leq 2^s d^{\frac{s}{2}} \widetilde{M},$$

which means that as long as we take $M \geq 2^s d^{\frac{s}{2}} \widetilde{M}$, we have

$$\mathcal{E}(M) \leq \inf_{f \in \Sigma_m^s} \|f - f_\rho\|_{L^\infty(\Omega)}^2 \lesssim \begin{cases} m^{-\frac{d+2s+1}{d}}, & \alpha \geq s + \frac{d+1}{2} \\ m^{-\frac{2\alpha}{d}}, & \alpha \in \mathbb{N} \cap (0, s + \frac{d+1}{2}) \end{cases}.$$

If $\alpha \geq s + \frac{d+1}{2}$, taking

$$M = 2^s d^{\frac{s}{2}} \asymp 1, \quad m \gtrsim n^{\frac{d}{2d+2s+1}}$$

yields

$$\mathcal{E}(M) \lesssim n^{-\frac{d+2s+1}{2d+2s+1}}, \quad \delta_n^2 \asymp n^{-\frac{d+2s+1}{2d+2s+1}} \log n, \quad \lambda = \frac{8\delta_n^2}{M} \asymp n^{-\frac{d+2s+1}{2d+2s+1}} \log n.$$

If $\alpha \in \mathbb{N} \cap (0, s + \frac{d+1}{2})$, taking

$$M = 2^s d^{\frac{s}{2}} m^{\frac{2s+d+1-2\alpha}{2d}}, \quad m \gtrsim n^{\frac{d}{2\alpha+d}}$$

yields

$$\mathcal{E}(M) \lesssim n^{-\frac{2\alpha}{2\alpha+d}}, \quad \delta_n^2 \asymp n^{-\frac{2\alpha}{2\alpha+d}} \log n, \quad \lambda = \frac{8\delta_n^2}{M} \asymp n^{-\frac{1}{2} - \frac{2s+1}{4\alpha+2d}} \log n. \quad (11)$$

In both cases, we have $\mathcal{E}(M) \lesssim \lambda M$, hence $\|f_m\|_{\mathcal{P}_{1,s}} \lesssim M$. Taking $\mathcal{F} = \text{star}(\pi_{2B}(\mathcal{B}_s(\|f_m\|_{\mathcal{P}_{1,s}} + M)))$ in Theorem 20, we obtain

$$\|\pi_{2B}(f_n - f_\rho)\|_{L^2(\mathrm{d}\mu)}^2 \lesssim \|\pi_{2B}(f_n - f_\rho)\|_{L^2(\mathrm{d}\mu_n)}^2 + \varepsilon_n^2 \quad \forall f \in \Sigma_{m,M}^s$$

with probability at least $1 - c_1 e^{-\frac{c_2 n \varepsilon_n^2}{B^2}}$, where

$$\varepsilon_n^2 \asymp n^{-\frac{d+2s+1}{2d+2s+1}} M^{\frac{2d}{2d+2s+1}} \log n M \asymp \delta_n^2.$$

Combining everything above, we obtain

$$\|\pi_B f_n - f_\rho\|_{L^2(\mathrm{d}\mu)}^2 \lesssim \mathcal{E}(M) + \delta_n^2 + \varepsilon_n^2,$$

with probability at least $1 - c_1 e^{-c_2 n \delta_n^2}$. The desired conclusion follows from (1) and (11). This completes the proof of Theorem 8. $\blacksquare$

Proofs of Theorems 11 and 12

To estimate the minimax lower bound, we first need some preliminaries. For each probability distribution ρ on $\Omega \times \mathbb{R}$, we can define a regression function

$$f_\rho(x) = \int_{\mathbb{R}} y \mathrm{d}\rho(y|x).$$

Under our assumptions, we consider only those ρ such that $f_\rho \in W^{\alpha,\infty}(\Omega)$, and denote the collection of all such ρ by $\mathcal{P}$. Let $\mathcal{F} = \{f_\rho : \rho \in \mathcal{P}\} \subset W^{\alpha,\infty}(\Omega)$. We say that $\{f_1, \dots, f_T\} \subset \mathcal{F}$ is a 2δ -separated set if for any $i \neq j$,

$$\|f_i - f_j\|_{L^2(\mathrm{d}\mu)} \geq 2\delta.$$

Write $f_i = f_{\rho_i}$, where $\rho_i \in \mathcal{P}$. We consider the *hypothesis testing problem* generated by the family of distributions $\{\rho_1, \dots, \rho_T\}$:

1. Select an index $J = j$ uniformly at random from the index set $[T] = \{1, \dots, T\}$.
2. Given $J = j$, draw a random sample Z from ρ_j , that is, $(Z|J = j) \sim \rho_j$.

Let $\bar{\rho}$ denote the joint distribution of (Z, J) . Note that the marginal distribution of Z is $\bar{\rho}_Z = \frac{1}{T} \sum_{j=1}^T \rho_j$. Given a sample Z , the hypothesis testing problem asks us to determine the random index J from this sample. A *test function* is a mapping $\Psi : \mathcal{Z} \rightarrow [T]$, and its error probability is defined by $\bar{\rho}(\Psi(Z) \neq J)$. The hypothesis testing problem is closely related to the minimax lower bound, namely (Wainwright, 2019, Proposition 15.1),

$$\inf_{\hat{f}} \sup_{f_\rho \in \mathcal{F}} \mathbb{E} \left[\|\hat{f} - f_\rho\|_{L^2(\mathrm{d}\mu)}^2 \right] \geq \delta^2 \inf_{\Psi} \bar{\rho}(\Psi(Z) \neq J),$$

where the infimum on the right-hand side is taken over all test functions. The difficulty of the hypothesis testing problem depends on the dependence between Z and J . In the most extreme case, Z and J are completely independent, in which case the observed sample Z is useless for determining J .

To measure the dependence between two probability measures, we introduce some information-theoretic concepts. Let $\mathbb{P}, \mathbb{Q}$ be probability measures on a measurable space $\mathcal{X}$ with base measure ν . Since they always admit densities with respect to the base measure $\nu = \frac{1}{2}(\mathbb{P} + \mathbb{Q})$, we may assume without loss of generality that their density functions are p, q , respectively. The commonly used *Kullback–Leibler divergence* in information theory is defined by

$$D(\mathbb{P} \parallel \mathbb{Q}) = \int_{\mathcal{X}} p(x) \log \frac{p(x)}{q(x)} d\nu(x),$$

which can to some extent measure the distance between two probability measures, although it is not a metric, since one can easily verify that it does not satisfy symmetry: $D(\mathbb{P} \parallel \mathbb{Q}) \neq D(\mathbb{Q} \parallel \mathbb{P})$. We know that two random variables (Z, J) are independent if and only if their joint distribution $\bar{\rho}$ equals

the product of their marginals $\bar{\rho}_Z \otimes \bar{\rho}_J$. Thus, it is natural to quantify the dependence between Z and J via the Kullback–Leibler divergence. Define the *mutual information* of (Z, J) by

$$I(Z; J) = D(\bar{\rho} \| \bar{\rho}_Z \otimes \bar{\rho}_J).$$

By the properties of the Kullback–Leibler divergence, we have $I(Z; J) \geq 0$, and $I(Z; J) = 0$ if and only if Z and J are independent.

Lemma 21 (Fano, (Wainwright, 2019, Proposition 15.12)) *The minimax risk of the function class $\mathcal{F}$ satisfies*

$$\inf_{\hat{f}} \sup_{f_\rho \in \mathcal{F}} \mathbb{E} \left[\|\hat{f} - f_\rho\|_{L^2(d\mu)}^2 \right] \geq \delta^2 \left(1 - \frac{I(Z; J) + \log 2}{\log T} \right).$$

By Fano’s lemma, it suffices to find an upper bound on $I(Z; J)$ in order to obtain a minimax lower bound. For this purpose, we can use the Yang–Barron inequality.

Lemma 22 (Yang–Barron, Yang and Barron (1999)) *Let $\mathcal{N}_{KL}(\varepsilon; \mathcal{P})$ be the ε -covering number of $\mathcal{P}$ under the square-root KL divergence. Then*

$$I(Z; J) \leq \inf_{\varepsilon > 0} \{ \varepsilon^2 + \log \mathcal{N}_{KL}(\varepsilon; \mathcal{P}) \}.$$

Combining Fano’s lemma and the Yang–Barron inequality, we can derive the minimax lower bound in two steps:

Proof of Theorem 11. Let $BW^{\alpha, \infty}(\Omega)$ denote the unit ball in $W^{\alpha, \infty}(\Omega)$. When $\mu \sim U(\Omega)$, we have

$$\log \mathcal{N}(\delta, BW^{\alpha, \infty}(\Omega), \|\cdot\|_{L^2(d\mu)}) \asymp \log \mathcal{N}(\delta, BW^{\alpha, \infty}(\Omega), \|\cdot\|_{L^2(\Omega)}) \asymp \delta^{-\frac{d}{\alpha}}.$$

Here the second equivalence follows from the Birman–Solomjak theorem (Birman and Solomyak, 1967). Therefore, we may choose a δ -separated set $\{f_1, \dots, f_T\}$ in $W^{\alpha, \infty}(\Omega)$ such that $\log T \gtrsim \delta^{-\frac{d}{\alpha}}$.

For $j \in [T]$, let ρ_j denote the distribution of the random vector y conditional on $\{\mathbf{x}_i\}_{i=1}^n$ when the true function is f_j , namely,

$$y_i = f_j(\mathbf{x}_i) + \eta_i, \quad \eta_i \sim \mathcal{N}(0, \sigma^2), \quad y = (y_1, \dots, y_n)^\top.$$

Let μ^n denote the joint distribution of $\{\mathbf{x}_i\}_{i=1}^n$. Then the joint distribution of $(y, \{\mathbf{x}_i\}_{i=1}^n)$ is $\rho_j \otimes \mu^n$. By the formula for the Kullback–Leibler divergence between Gaussian measures, for any $j \neq k$,

$$D(\rho_j \otimes \mu^n \| \rho_k \otimes \mu^n) = \mathbb{E}_x [D(\rho_j \| \rho_k)] = \frac{n}{2\sigma^2} \|f_j - f_k\|_{L^2(d\mu)}^2.$$

Therefore,

$$\log \mathcal{N}_{KL}(\varepsilon; \mathcal{P}) \leq \log \mathcal{N} \left(\varepsilon \sqrt{\frac{2\sigma^2}{n}}, BW^{\alpha, \infty}(\Omega), \|\cdot\|_{L^2(d\mu)} \right) \lesssim \left(\frac{\varepsilon^2}{n} \right)^{-\frac{d}{2\alpha}}.$$

Combining Fano’s lemma and the Yang–Barron inequality, we can prove the conclusion by exactly the same argument as in the previous theorem. This completes the proof of Theorem 11. $\blacksquare$

The proof of Theorem 12 follows by exactly the same method, and is therefore omitted.

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